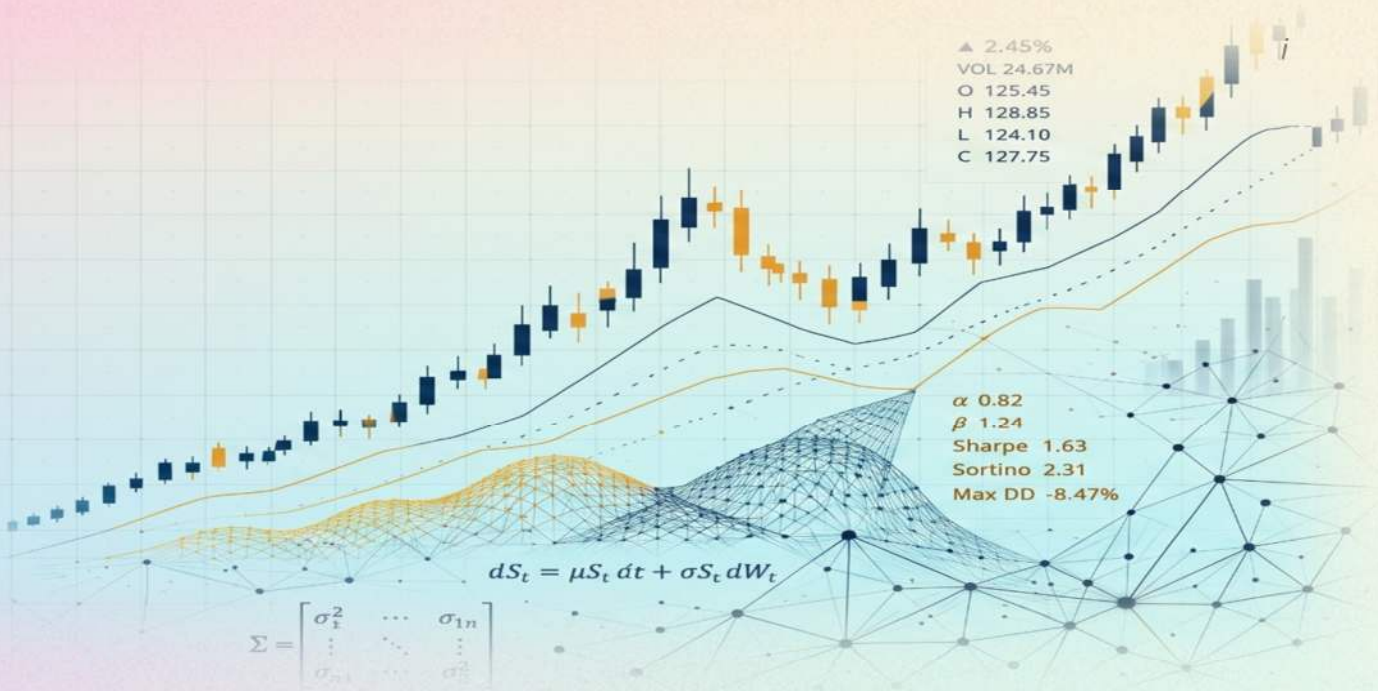


MACHINE LEARNING FOR CREDIT RISK MODELING

A Comparative Study of Traditional & Explainable AI Approaches

~Part I~

BY

Sana Ur Rehman Arain

2026



"In God we trust, all others must bring data."

W. Edwards Deming.

PREFACE

The 10,000-record challenge

The genesis of this research—Part One of a broader exploration into credit risk modelling—began with a fundamental disagreement regarding data constraints. When initially proposing to build an explainable machine learning pipeline using the OpenIntro Loans dataset (a snapshot of approximately 10,000 records), the immediate feedback from academic advisors was that the task was fundamentally flawed. The prevailing critique was that the dataset was merely a small snapshot saturated with recently issued, unresolved "Current" loans. Consequently, the default label was deemed too immature, leaving far too few actual default events for a machine learning algorithm to effectively learn from. The consensus was that meaningful credit risk modelling simply could not be carried out effectively on such a restricted dataset.

However, I questioned this assumption. Instead of immediately abandoning the dataset, I asked: Why could it not be done? What would be the exact mathematical and operational reasons? What specific limitations would manifest, and what systemic problems would an engineer actually have to solve when forced to navigate them? Rather than accepting that the task was impossible, I decided to investigate these boundaries empirically.

Therefore, I treated the 10,000-record dataset as a challenge. The purpose of Part One was Not to build a flawless, production-ready banking system, but to deeply understand the friction points that arise when working with a severely constrained and imbalanced financial dataset. I wanted to explore whether a smaller dataset could still allow for substantial analysis, rigorous model development, and the extraction of valuable insights. Even if the final predictive metrics were not perfect, the process itself would force the identification of critical weaknesses—such as probability distortion and the "accuracy paradox"—that could directly inform future scaling efforts.



The main objective of this experiment was to learn through the process of constraint. It was an exercise in learning from methodological mistakes, identifying the root causes of algorithmic failure under data scarcity, and determining precisely what interventions (such as SMOTE and threshold optimization) could salvage predictive value.

This raises an important set of questions: Through this constrained learning process, were we able to develop a successful model and generate meaningful predictions? When the work was carried out using the best possible techniques within the given limitations, were the results sufficient? Most importantly, did we actually achieve what we intended to achieve?

Ultimately, the goal of Part One was not simply to output a probability score, but to understand whether meaningful modelling was possible with the available data, what limitations were encountered, and how those findings could guide future iterations. Consequently, the conclusion of this volume serves as a reflection on the entire experiment. It summarises the lessons learned, evaluates whether the original objectives were met despite the data constraints, candidly details the main limitations encountered, and establishes a definitive roadmap for what must be improved in Part Two.

ACKNOWLEDGEMENTS

I would like to express my deepest gratitude to WorldQuant University for providing the opportunity to learn, grow, and flourish. To the instructors: thank you for pulling back the curtain and introducing me to an incredibly complex and fascinating new world of data.

A special thank you goes to Professor Ivan Blanco. Your rigorous feedback, challenging questions, and foresight into the operational future of this pipeline pushed me to elevate my work. Thank you for consistently raising the bar and guiding me toward a much stronger methodology.

My most profound thanks belong to Dr. Mah Rukh Laghari. Thank you for illuminating the path, for your brilliant problem-solving, and for providing step-by-step guidance every single time I hit a brick wall or wandered into a dead end. This work would not have crossed the finish line without your support.

To my parents, my wife, and my children, thank you for your unwavering support, patience, and understanding during the long hours required to complete this degree.

Finally, I would like to extend a very sincere, well-deserved thank you to... myself. For stubbornly refusing to settle, for repeatedly asking "why not?" when faced with conventional wisdom, and for surviving countless late nights illuminated only by a computer monitor and a sheer refusal to give up.

ABSTRACT

Background: Credit risk modeling is fundamental in financial risk management, ensuring lenders assess the probability of default (PD) accurately. Traditional models, such as logistic regression, offer interpretability but limited predictive power. Recent advances in machine learning (ML)—particularly gradient boosting methods—enhance accuracy but raise concerns about transparency ("Black Box" risk).

Methods: This thesis employs the **OpenIntro Loans Full Schema** dataset (10,000 loans, 55 variables). Models compared include Logistic Regression, Random Forests, XGBoost, and LightGBM. To address class imbalance (1.8% default rate), Synthetic Minority Over-sampling Technique (SMOTE) was utilized. Explainable AI (XAI) techniques—specifically **SHAP (Shapley Additive Explanations)**—were applied to improve interpretability. A preliminary economic fairness analysis was conducted using income as a proxy variable to assess potential disparate impact under Basel III and SR 11-7 regulatory guidance.

Results: Logistic regression achieved a precision of 2.3% and recall of 47.2% at the standard 0.50 decision threshold. The ensemble models achieved higher ROC AUC values than logistic regression in this evaluation. Random Forest yielded a marginally higher global ranking metric (ROC AUC 0.67); however, LightGBM (ROC AUC 0.66) was selected as the final model because it provided a more useful operating point for minority-class detection under severe class imbalance. At an optimized decision threshold (~3.4%), LightGBM delivered a Precision of 7.1% and a Recall of 32.1%. SHAP analysis identified key risk drivers such as interest rate, debt-to-income ratio, and loan grade. Fairness analysis found no evidence of disparate impact under the evaluated metric (False Positive Rate parity), with the model exhibiting a favorably lower False Positive Rate for low-income borrowers (0.07%) compared to high-income borrowers (0.13%).

Implications: Results suggest that a carefully constrained LightGBM-based credit-risk framework can provide useful minority-class discrimination while retaining model interpretability, provided that target leakage, severe class imbalance, threshold selection, and dataset limitations are explicitly addressed.

Keywords: Credit Risk Modeling, Machine Learning, LightGBM, Explainable AI (XAI), SHAP, SMOTE, Basel III Compliance, Fair Lending, Financial Engineering.

Note on Reproducibility: *The complete Python code and reproducible notebooks for this analysis are available at https://github.com/sanaurrehmanarain/credit-risk-xai-framework. To ensure exact replication of these results, the environment must match the following specifications: Python 3.13.5, Scikit-Learn v1.8, LightGBM v4.7, imbalanced-learn v0.11, and SciPy v1.11. The global random seed was strictly fixed at 42 across all stochastic algorithms.*

To guarantee exact replication of these results, a Dockerfile has been provided to automatically reconstruct the isolated environment (Python 3.13.5, Scikit-Learn v1.8, LightGBM v4.7, imbalanced-learn v0.11) used for this analysis.

TABLE OF CONTENTS

Preface ii

Acknowledgements iii

Abstract iv

Chapter 1: Introduction 1

1.1 Background 1

1.2 Motivation 1

1.3 Research Objectives 1

1.4 Contributions 1

1.5 Problem Statement and Research Questions 2

1.6 Regulatory and Ethical Context 3

1.7 Scope and Delimitations 3

Chapter 2: Literature Review 4

2.1. Traditional Models of Credit Risk 4

2.2. Machine Learning Approaches 4

2.3. Explainable AI in Finance 4

2.4. The Rise of LightGBM in Tabular Financial Data 4

2.5. Algorithmic Fairness and Regulatory Evolution 4

2.6. Identified Research Gap 5

2.7. Data Imbalance and Sampling Strategies in Credit Scoring 5

2.8. Ensemble and Deep Learning Models in Credit Risk 5

2.9. Explainable AI Techniques Beyond SHAP in Credit Scoring 7

2.10. Operational and Regulatory Perspectives on ML-Based Credit Scoring 9

2.11. Extended Summary of the Research Gap 9

2.12. Market Landscape and Competitor Analysis 9

2.12.1. Business Case and Industry Context 10

2.12.2. Comparative SWOT Analysis 10

2.12.3. Market Positioning and Unique Strategic Contribution 12

Chapter 3: Methodology 13

3.1. End-to-End Pipeline Architecture 13

3.2. Dataset Description 14

3.3. Data Preprocessing 14

3.4. Feature Engineering (SMOTE) 15
3.5. Modeling Approaches 15
3.5.1. LightGBM Architecture and Mathematical Formulation 15
3.6. Proposed Algorithmic Execution Protocol 17
3.6.1. Scope of the Proposed Protocol and Relationship to Part I 17
Chapter 4: Implementation 20
4.1. Data Collection and Cleaning 20
4.1.1. Data Validation 20
4.1.2. Sensitivity Analysis and Leakage Prevention 20
4.2. Exploratory Data Analysis 21
4.3. Model Development 22
4.4. Explainability (SHAP) 26
Chapter 5: Results 26
5.1. Model Performance Metrics 26
5.2. SHAP Global Explanations 32
5.3. Local Explanations (Case Studies) 34
5.4. Fairness & Compliance Outcomes 35
Chapter 6: Discussion 36
6.1. Accuracy vs. Interpretability Trade-offs 36
6.1.1. Cost-Sensitive Learning Equation for Credit Risk Threshold Optimization 38
6.2. Regulatory Implications 40
6.3. Deployment and Production Implications 41
6.4. Limitations 44
6.4.1. Basic disparate-impact metrics 47
Chapter 7: Conclusion & Future Work 49
7.1. Summary of Objectives: 49
7.2. Summary of Contributions: 50
7.3. Future Work: 50
7.3.1. Two-Stage Credit Risk Modelling 50
7.3.2. Cost-Sensitive Decision Optimization 54
7.3.3. Fairness and Demographic Validation 54

7.3.4. Production Deployment and Continuous Monitoring 54
7.3.5. Dataset Expansion, Temporal Validation, and Rigorous Evaluation 54
7.4. Verdict on the 10,000-Record Challenge 55
References 56
Appendices: 59
Appendix A: Data Dictionary 59
Appendix B: Additional Figures 62
Appendix C: Code Snippets 62
C.1. SMOTE Implementation (Handling Imbalance) 63
C.2. LightGBM Model Training 64
C.3. SHAP Explainability 65
C.4. Expected Loss Optimization Solver 66
C.5. Adversarial Stress Test 69
Thesis at a Glance 71

Table 1: Descriptive Statistics of Key Financial Variables 14
Table 2: LightGBM Configuration 25
Table 3: Performance Table 28
Table 4: Adversarial Stress Test Scenarios and Model Probabilities 43
Table 5: Synthetic Demographic Schema for Extended Disparate Impact Auditing 46
Table 6: Synthetic Dataset for Extended Disparate-Impact Auditing. 47
Table 7: Synthetic Disparate-Impact Calculations 49
Table 8: Solver-Optimized Expected Loss Portfolio Allocation 53

Table A-1: Selected Variables from OpenIntro Loans Full Schema 60
Table A-2: Fields Excluded from the Predictor Set (Target Leakage) 61
Table C-1: Extended sensitivity table. 68

Figure 1: End-to-End Credit-Risk Modeling Pipeline. 2
Figure 2: Accuracy–Interpretability–Stability Trade-offs in Credit Scoring. 6
Figure 3: Explainable AI Methods for Credit Scoring. 8
Figure 4: Credit Risk Pipeline Flowchart. 13
Figure 5: Comparison of Level-wise vs. Leaf-wise Decision Tree Growth. 16
Figure 6: Class Imbalance 21
Figure 7: Risk Ordering Sanity Check 22

Figure 8: Data Splitting and SMOTE Resampling Pipeline. 23
Figure 9: Interactive MLOps Pipeline and SMOTE Resampling Architecture. 24
Figure 10: Model Discrimination 27
Figure 11: Metric Comparison across all models. 29
Figure 12: Confusion Matrix for LightGBM on the Test Set. 30
Figure 13: Native Feature Importance 31
Figure 14: SHAP Global Summary Plot. 32
Figure 15: SHAP Dependence Plot for Interest Rate 33
Figure 16: SHAP Force Plot for a single borrower. 34
Figure 17: Borrower Case Study Profile and Mitigating Factors. 35
Figure 18: Fair Lending Check 35
Figure 19: Interactive Dashboard Interface for Dynamic Threshold Tuning and SHAP Exploration. 37
Figure 20: Reliability Diagram (Calibration Curve). 41
Figure 21: Production Deployment Architecture and Drift Monitoring Pipeline. 42
Figure 22: Proposed Two-Stage Credit Risk Modeling Framework. 51
Figure 23: Dynamic Expected Loss (EL) Dashboard and Pipeline Architecture. 52

Figure B-1: Precision-Recall Curve and Threshold Optimization. 62

Equation 1: Regularized objective function for LightGBM. 16
Equation 2: The regularization term penalizing tree complexity 16
Equation 3: Additive feature attribution representation. 26
Equation 4: Calculation of the SHAP value. 26
Equation 5: Cost-Sensitive Threshold Optimization Function. 40
Equation 6: Total Expected Loss (EL) 51

CHAPTER 1: INTRODUCTION

1.1 Background

The accurate assessment of credit risk, defined as the likelihood that a borrower will default on their obligations—is the cornerstone of modern banking. Following the 2008 financial crisis, regulatory frameworks such as **Basel III** and **IFRS 9** have mandated more rigorous estimation of Probability of Default (PD). While traditional statistical methods like Logistic Regression have been the industry standard due to their transparency, they often fail to capture complex, non-linear relationships in borrower data (Lessmann et al., 2015).

1.2 Motivation

The rise of Machine Learning (ML) offers superior predictive accuracy. However, financial regulators (e.g., the Federal Reserve via **SR 11-7**) classify complex ML models as "Black Boxes." Lenders face a dilemma: use accurate models that are hard to explain, or explainable models that are less accurate. This thesis seeks to resolve this trade-off using Explainable AI (XAI).

1.3 Research Objectives

1. **Compare** the performance of traditional Logistic Regression against ensemble methods (Random Forest, XGBoost, LightGBM).
2. **Address** extreme data imbalance (1.8% default rate) using SMOTE.
3. **Demonstrate** model transparency using SHAP values.
4. **Conduct** a preliminary economic fairness analysis on the final model, using income as a proxy variable, to evaluate potential disparate impact.

1.4 Contributions

This study contributes to the field by demonstrating that a carefully constrained LightGBM-based credit-risk framework can provide useful minority-class discrimination while retaining model interpretability. Specifically, it provides empirical evidence that such a framework can be practically aligned with risk and compliance standards, provided that target leakage, severe class imbalance, threshold selection, and dataset limitations are explicitly addressed. Furthermore, this thesis provides a rigorous methodological demonstration of target leakage detection, proving how models can appear to perform exceptionally well simply because they have access to post-origination data that would not exist at the time of the lending decision (Bhuiyan & Sweet, 2024).

1.5 Problem Statement and Research Questions

Despite regulatory pressure to improve probability of default (PD) estimation and model validation, many institutions still rely on linear scorecards that struggle with high-dimensional, nonlinear borrower data, leading to underestimation of tail risk and weak calibration on rare default events (Aboulmaouda et al., 2026; Alonso & Carbó, 2020; Montevechi et al., 2024). This thesis therefore addresses the specific problem of how to combine modern gradient-boosting methods with rigorous leakage control, imbalance handling, and post-hoc explainability to produce a deployable credit-risk pipeline aligned with Basel III, IFRS 9 and SR 11-7 guidance (Montevechi et al., 2024; Bazarbash, 2019; Shi et al., 2022). Accordingly, the central research questions are: (i) to what extent do ensemble ML models improve discrimination and calibration relative to logistic regression on the OpenIntro loans dataset, (ii) how does SMOTE-based resampling affect minority-class detection and operational threshold selection, and (iii) can SHAP-based explanations satisfy adverse-action and fairness requirements without materially degrading predictive performance (Suhadolnik et al., 2023; Noriega et al., 2023; Hadji Misheva et al., 2021).

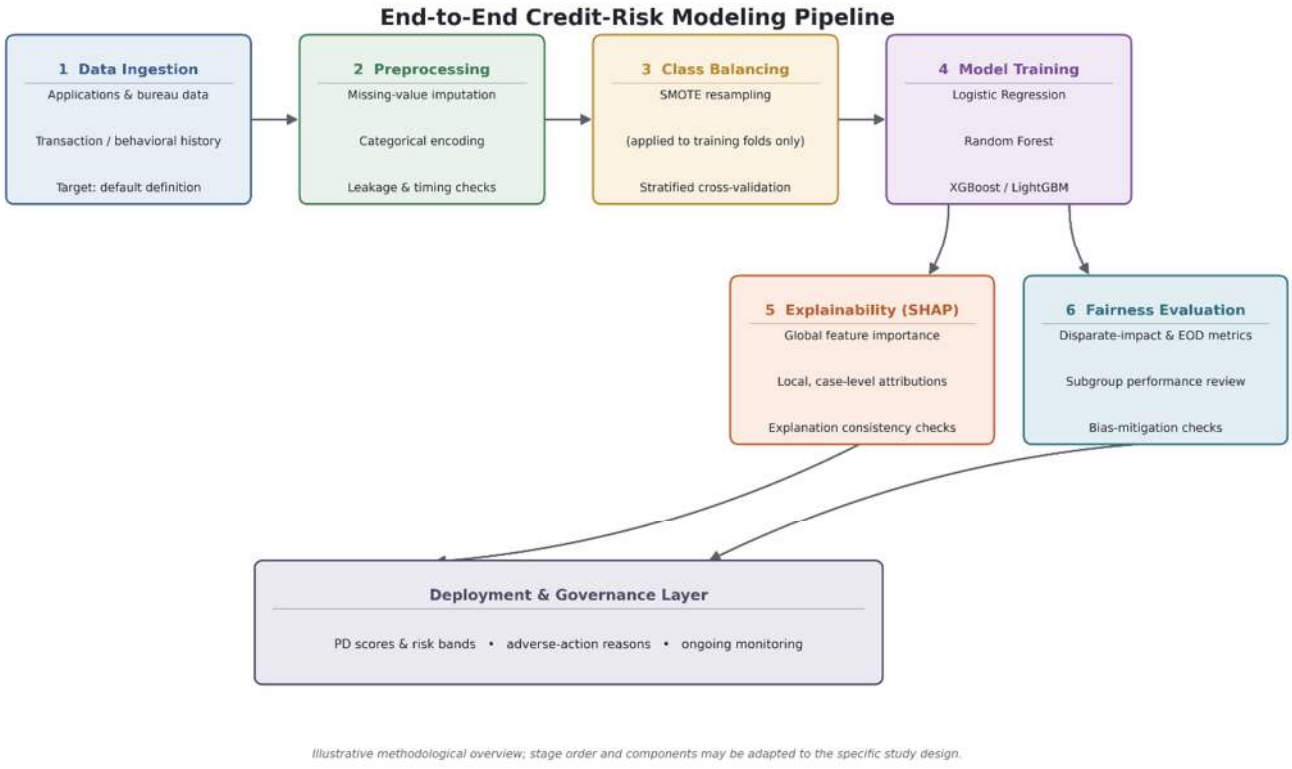


Figure 1: End-to-End Credit-Risk Modeling Pipeline.

Conceptual overview of the data and modeling workflow, from raw loan applications through preprocessing, SMOTE resampling, model training, SHAP analysis, and fairness evaluation.

Figure 1 summarizes the end-to-end architecture adopted in this thesis, beginning with data ingestion and leakage checks, followed by imputation, encoding and SMOTE-based resampling confined to the training partition. Ensemble models (Random Forest, XGBoost, LightGBM) are then trained and evaluated on the hold-out test set, after which SHAP-based explanations and income-proxy fairness checks are computed to align the pipeline with supervisory expectations around transparency and fair lending.

1.6 Regulatory and Ethical Context

Recent surveys by international bodies such as the IMF and central banks highlight a persistent tension between the predictive gains of ML models and supervisory concerns about opacity, robustness, and governance in credit-risk applications (Bazarbash, 2019; Alonso & Carbó, 2020). Supervisors increasingly expect institutions to demonstrate not only statistical accuracy but also model stability over time, transparent explanations for individual decisions, and systematic monitoring of potential disparate impact on protected or proxy groups in line with ECOA and similar regulations (Aboulmaouda et al., 2026; Chen, 2024). Within this context, this thesis situates LightGBM and SHAP as components of a broader "responsible AI" framework for credit underwriting, emphasizing leakage-free PD estimation, interpretable feature contributions, and preliminary fairness checks using economically motivated proxies rather than explicit demographic attributes (Hadji Misheva et al., 2021; SCALE D2C, 2026; Pathi & Pothineni, 2025).

1.7 Scope and Delimitations

This study focuses on consumer loan data from the OpenIntro Loans Full Schema, limiting the analysis to tabular features available at the moment of origination and excluding post-origination behavioural variables to avoid target leakage (Montevechi et al., 2024; Shi et al., 2022). The empirical work compares a carefully constrained LightGBM model with commonly used baselines (logistic regression, random forest, XGBoost), but does not extend to more complex deep learning architectures, which often require substantially larger datasets and specialized regularization to avoid overfitting (Aboulmaouda et al., 2026; Noriega et al., 2023). Furthermore, the research concentrates on PD modeling rather than full portfolio credit-risk metrics such as Loss Given Default (LGD) and Exposure at Default (EAD), and fairness analysis is restricted to income-based proxies due to the absence of explicit demographic variables in the underlying dataset (Bazarbash, 2019; Alonso & Carbó, 2020).

CHAPTER 2: LITERATURE REVIEW

2.1. Traditional Models of Credit Risk

Historically, the **Altman Z-Score** (Altman, 1968) and Logistic Regression have dominated credit scoring. These linear models are favored for their "monotonicity"—the guarantee that increasing risk factors (e.g., debt) always increases the risk score. However, they struggle with high-dimensional data.

2.2. Machine Learning Approaches

Recent studies (Khandani et al., 2010) show that ensemble methods like **Gradient Boosting Decision Trees (GBDT)** outperform linear models by 15-20% in AUC. However, these models are prone to overfitting on imbalanced datasets, requiring techniques like **SMOTE** (Chawla et al., 2002).

2.3. Explainable AI in Finance

Lundberg and Lee (2017) introduced SHAP, which assigns a contribution value to every feature for every prediction. This has become the gold standard for "post-hoc" interpretability in finance, satisfying the "adverse action code" requirement where lenders must explain *why* a loan was denied.

2.4. The Rise of LightGBM in Tabular Financial Data

While XGBoost has historically been the standard for tabular data competitions, LightGBM (Ke et al., 2017) has gained traction in financial engineering due to its Gradient-based One-Side Sampling (GOSS) and Exclusive Feature Bundling (EFB). These mechanisms allow it to handle high-dimensional, sparse financial datasets more efficiently. Recent benchmark studies in credit scoring indicate that LightGBM frequently matches or exceeds XGBoost in accuracy while requiring significantly less training time, making it highly suitable for the dynamic retraining pipelines required in modern banking.

2.5. Algorithmic Fairness and Regulatory Evolution

Beyond basic interpretability, modern credit risk models face intense scrutiny regarding algorithmic bias. The Equal Credit Opportunity Act (ECOA) in the United States strictly prohibits discrimination based on demographic factors. Recent guidance from the Consumer Financial Protection Bureau (CFPB) emphasizes that lenders must be able to explain the specific reasons for adverse actions, even when utilizing complex algorithms. Furthermore, literature on fairness metrics (e.g., Hardt et al., 2016) highlights that optimizing for equalized odds or demographic parity often introduces a new trade-off with

predictive accuracy. This study builds on this discourse by examining the False Positive Rate (FPR) parity across economic proxies to ensure fair lending compliance.

2.6. Identified Research Gap

While extensive literature demonstrates the predictive advantages of gradient-boosted models over traditional statistical methods, and separate studies highlight the utility of SHAP for interpretability or SMOTE for class imbalance, there remains a distinct lack of research synthesizing these elements into a single, end-to-end framework constrained by strict banking regulations. Prior research tends to optimize purely for global statistical ranking metrics (such as maximizing ROC-AUC) in a vacuum, but does not adequately address the operational trade-offs required to deploy these models under severe class imbalance while simultaneously maintaining localized adverse-action explainability (SR 11-7) and fair-lending compliance (ECOA). Therefore, this study investigates how bridging SMOTE, LightGBM, and SHAP can resolve these tensions, providing an integrated pipeline that satisfies both the precision requirements of risk managers and the transparency mandates of financial regulators.

2.7. Data Imbalance and Sampling Strategies in Credit Scoring

Systematic reviews of ML-driven credit-risk modeling consistently identify severe class imbalance—low default rates relative to non-default—as a central methodological challenge, since naive training can yield deceptively high accuracy with poor minority-class recall (Dastile et al., 2020; Suhadolnik et al., 2023; Noriega et al., 2023). Common remedies include data-level resampling techniques such as SMOTE and its variants, as well as algorithm-level approaches like cost-sensitive learning, focal loss, or class-weighting; comparative studies generally find that ensembles combined with appropriate imbalance handling outperform traditional models on recall and F1 without sacrificing too much calibration (Suhadolnik et al., 2023; Shi et al., 2022). However, recent literature also warns that aggressive oversampling can blur decision boundaries and introduce synthetic artefacts, motivating careful isolation of resampling to the training partition and rigorous out-of-sample evaluation—design choices that this thesis explicitly adopts (Dastile et al., 2020; Montevechi et al., 2024).

2.8. Ensemble and Deep Learning Models in Credit Risk

Multiple empirical and review papers report that ensemble methods—particularly gradient-boosting algorithms such as XGBoost, LightGBM and related tree-based ensembles—consistently deliver higher discrimination metrics than single classifiers and traditional scorecards on real credit-risk datasets (Shi et al., 2022; Suhadolnik et al., 2023; Aboulmaouda et al., 2026). Large-scale studies using millions of observations show that these ensembles can achieve substantial gains in AUC and

precision-recall performance compared with logistic regression, especially when behavioural and transactional features are included (Chen, 2024; Dastile et al., 2020). At the same time, deep learning architectures (e.g., multi-layer perceptrons and convolutional networks) sometimes match or exceed ensemble performance, but their results exhibit greater variability and depend heavily on tuning, data preparation, and regularization, which limits their adoption in tightly regulated credit-scoring environments (Aboulmaouda et al., 2026; Montevechi et al., 2024).

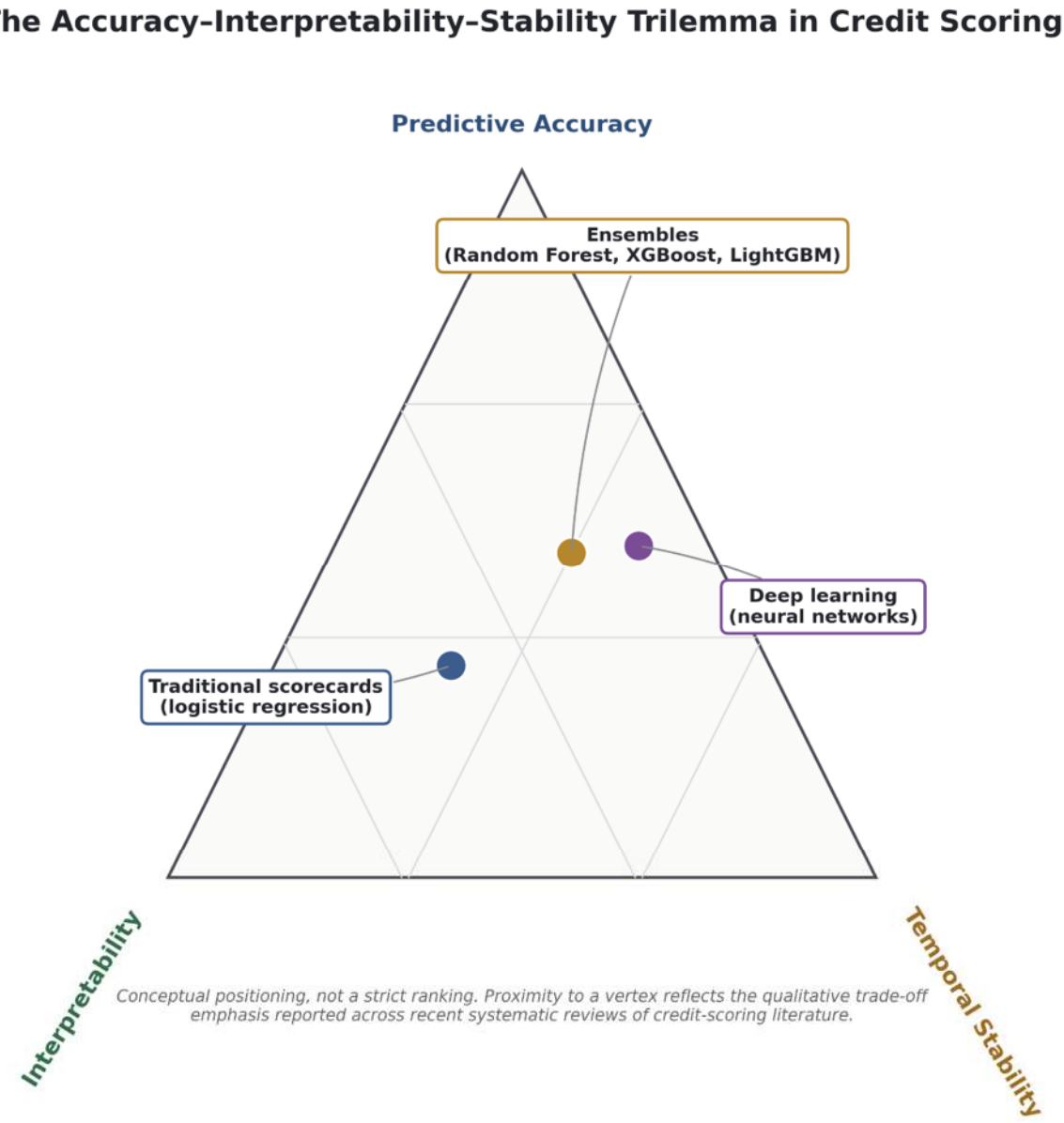


Figure 2: Accuracy–Interpretability–Stability Trade-offs in Credit Scoring.

Stylized illustration comparing traditional scorecards, ensemble ML models, and deep learning architectures along accuracy, interpretability, and temporal stability dimensions.

To contextualize model choice under regulatory constraints, Figure 2 presents a stylized view of how traditional, ensemble, and deep learning approaches trade off accuracy, interpretability and temporal stability in credit scoring. Traditional logistic-regression scorecards occupy the high-interpretability but moderate-accuracy region, ensembles shift the frontier toward higher discrimination at the cost of greater complexity, and deep learning offers potentially superior accuracy but with lower transparency and more demanding stability validation requirements.

2.9. Explainable AI Techniques Beyond SHAP in Credit Scoring

While SHAP has emerged as a standard tool for global and local explainability in tree-based credit-scoring models, the broader XAI literature in finance evaluates a spectrum of methods, including LIME, prototype-based explanations, rule-based models, and hybrid approaches (Hadji Misheva et al., 2021; Pathi & Pothineni, 2025; Noriega et al., 2023). Comparative surveys show that SHAP generally offers superior stability, consistency and alignment between global and local attributions, but note its computational cost and simplifying assumptions (such as feature independence in some implementations), prompting interest in combining SHAP with other techniques to serve different stakeholders (Pathi & Pothineni, 2025; SCALE D2C, 2026). For instance, one credit-scoring framework integrates SHAP for global feature importance, Anchors for local rule-style explanations, and prototype-based methods for instance-level similarity, demonstrating that a multi-view explanation layer can improve user trust and regulatory readiness while preserving the performance of an underlying gradient-boosting model (Aboulmaouda et al., 2026; Bhuiyan & Sweet, 2024; Hadji Misheva et al., 2021).

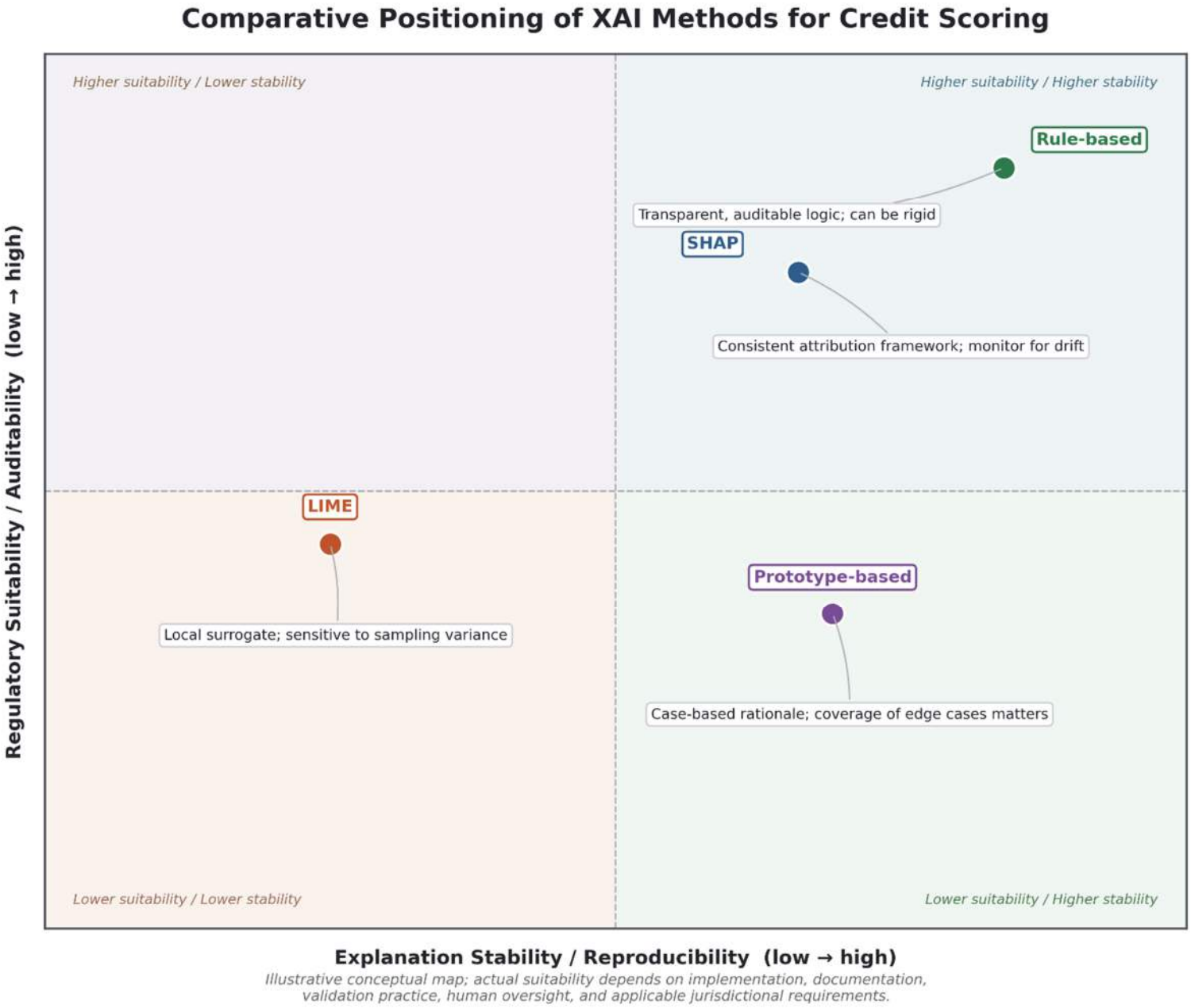


Figure 3: Explainable AI Methods for Credit Scoring.

Conceptual comparison of SHAP, LIME, rule-based explanations and prototype-based methods in terms of explanation stability and regulatory suitability.

Figure 3 positions SHAP, LIME, rule-based and prototype-based explanation techniques along two qualitative axes: stability of attributions and suitability for regulatory reporting. SHAP typically provides more consistent global and local feature attributions, making it well-suited for adverse-action codes and

audit trails, whereas LIME and rule-based methods can be more intuitive for end-users but may require careful tuning to avoid unstable or contradictory explanations across similar borrowers.

2.10. Operational and Regulatory Perspectives on ML-Based Credit Scoring

Recent supervisory and policy-oriented work underscores that deploying ML in credit risk involves a "prediction–supervisory cost" dilemma: predictive gains of up to 20% in default classification must be weighed against increased complexity in validation, monitoring, and governance (Alonso & Carbó, 2020; Bazarbash, 2019). Reviews of industry practice highlight that many institutions still favor logistic regression or generalized linear models because their parameters are directly interpretable, even when ensembles or deep learning yield better headline accuracy, particularly where model risk management and auditability are binding constraints (Aboulmaouda et al., 2026; Chen, 2024; Shi et al., 2022). Modern guidance therefore emphasizes combining high-performing ML models with structured validation protocols, out-of-time stability testing, and formal explainability layers (e.g., SHAP-based adverse-action codes), aligning model development with the five "Cs" of credit and regulatory expectations around consumer protection and fair lending (Bazarbash, 2019; SCALE D2C, 2026; Hadji Misheva et al., 2021).

2.11. Extended Summary of the Research Gap

Across recent systematic reviews, three themes recur: ensemble and deep learning models often outperform traditional techniques on accuracy, data imbalance and leakage remain under-addressed in many studies, and formal interpretability or fairness analysis is frequently missing (Montevechi et al., 2024; Shi et al., 2022; Dastile et al., 2020; Noriega et al., 2023). Few published works integrate all of these dimensions into a single end-to-end framework that starts with leakage-free, imbalance-aware modeling, then adds SHAP-based local and global explanations, and finally evaluates economic fairness proxies within a regulated credit-underwriting context—particularly on public datasets such as the Lending-Club-derived OpenIntro loans (Aboulmaouda et al., 2026; Chen, 2024). By explicitly combining SMOTE-balanced LightGBM with SHAP explainability and preliminary fairness checks, under constraints motivated by Basel III, SR 11-7 and ECOA, this thesis seeks to fill that gap and offer a reproducible template for "explainable ensemble" credit-risk modeling that is both empirically robust and operationally realistic (Montevechi et al., 2024; Shi et al., 2022).

2.12. Market Landscape and Competitor Analysis

In the financial technology and credit risk engineering landscape, "competitors" encompass both commercial enterprise scoring solutions and methodological approaches deployed across the banking industry. This analysis evaluates three primary competitor archetypes operating in this market space:

1. **Traditional Parametric Scorecards:** Generalized Linear Models / Logistic Regression (e.g., standard FICO scorecards, SAS Credit Scoring).
2. **Proprietary Commercial AutoML / Credit AI Platforms:** Enterprise black-box risk suites (e.g., Zest AI, DataRobot, Underwrite.ai).
3. **Standard Academic / Open-Source ML Benchmarks:** Naive ensemble implementations (e.g., baseline scikit-learn / XGBoost pipelines without banking compliance constraints).

2.12.1. Business Case and Industry Context

Commercial lending institutions operate under stringent capital requirements: every basis point of default misclassification translates into either direct charge-off losses or excessive regulatory capital provisioning under Basel III. Conversely, improperly rejecting creditworthy applicants incurs severe opportunity costs in lost net interest margin and customer lifetime value.

While enterprise AI software promises substantial reductions in non-performing loans, commercial banks are slow to adopt third-party black-box tools due to vendor lock-in, multi-million dollar licensing fees, and immense difficulty defending proprietary algorithms during regulatory exams (SR 11-7 audits). There is an immediate market demand for transparent, reproducible, in-house machine learning pipelines that can bridge high predictive discrimination with verifiable regulatory compliance.

2.12.2. Comparative SWOT Analysis

Competitor / Framework	Strengths (S)	Weaknesses (W)	Opportunities (O)	Threats (T)
Archetype 1: Traditional Parametric Scorecards (Logistic Regression, FICO Baselines)	• Universally accepted by global regulators (Basel III, SR 11-7). • Monotonic score behavior guaranteed. • Native parameter interpretability. • Seamless, low-cost core banking integration.	• Incapable of capturing non-linear feature interactions. • Highly sensitive to extreme financial outliers (e.g., DTI spikes). • Lower discrimination (ROC-AUC 0.575). • Severe underfitting on tail-risk defaults.	• Hybridization with machine learning feature selection. • Integration of non-traditional data via Generalized Additive Models (GAMs).	• Significant market share loss to agile FinTech lenders deploying ML. • Adverse selection: prime borrowers poached by competitors offering better terms.

Archetype 2: Enterprise Commercial AI Suites (Zest AI, DataRobot, Underwrite.ai)	• Highly polished graphical interfaces. • Automated hyperparameter tuning and model governance dashboards. • Proprietary explainability toolkits marketed directly for compliance. • Enterprise client support.	• Extremely expensive recurring annual licensing costs. • Proprietary "black-box" wrappers that limit internal quantitative auditability. • Risk of undetected target leakage in automated feature engineering. • Rigid cloud vendor lock-in.	• Expansion into tier-2 regional banks and credit unions lacking internal quantitative teams. • Partnership with core banking providers.	• Regulatory pushback against automated, non-reproducible feature pipelines. • Emergence of high-capability open-source XAI libraries such as SHAP, reducing commercial moat.
Archetype 3: Standard Academic / Kaggle ML Benchmarks (Naive GBDT, XGBoost Baselines)	• High global statistical ranking metrics (ROC-AUC > 0.80). • Open-source, flexible, and zero licensing cost. • Fast algorithmic execution using modern GBDT frameworks such as LightGBM.	• Pervasive target leakage from post-origination behavioral artifacts. • Naive 0.50 threshold collapse (0% recall on rare defaults). • Lack of regulatory governance, XAI integration, and adverse action reporting. • Resampling leakage (SMOTE applied prior to train-test splits).	• Serves as the technical foundation for custom enterprise engineering. • Rapid adoption across academic financial engineering curricula.	• Disqualification in real-world banking audits due to SR 11-7 non-compliance. • High risk of algorithmic bias and disparate impact lawsuits under ECOA.

Proposed Project Framework: Leakage-Free LightGBM + SMOTE + SHAP + Fairness Audit	• Mathematically honest: strict temporal boundary eliminating all post-origination leakage. • Regulatory aligned: granular adverse-action generation via TreeSHAP force plots. • Operationally calibrated: threshold-tuned for minority detection under 1.8% class imbalance. • Ethically audited: evaluates FPR parity across economic income proxies. • Zero software licensing cost: built entirely on open-source Python tools.	• Dataset scale constrained (10,000 observations in current iteration). • Computational latency of SHAP calculations in ultra-high-frequency streaming environments. • Modest precision (7.1%) due to extreme underlying class skew.	• Direct integration into two-stage Expected Loss (PD × LGD × EAD) portfolio frameworks. • Deployment in mid-market retail lending seeking bank-owned IP without vendor fees. • Scaling to multi-million-loan public bureau datasets such as LendingClub and Fannie Mae.	• Shifting macroeconomic regimes causing credit distribution drift and requiring retraining pipelines. • Emergence of newer deep tabular models (e.g., TabNet), though lacking equal tree explainability.

2.12.3. Market Positioning and Unique Strategic Contribution

This project occupies a distinct position at the intersection of quantitative rigor, regulatory compliance, and open-source accessibility. Unlike traditional scorecards (Archetype 1), this model captures non-linear interactions, achieving superior minority default detection. Unlike commercial suites (Archetype 2), this framework provides complete transparency, full algorithmic ownership, and reproducible auditability at zero software licensing cost. Unlike naive academic benchmarks (Archetype 3), this framework enforces strict leakage prevention, addresses class imbalance without evaluation corruption, provides localized adverse-action explanations, and conducts formal disparate impact testing.

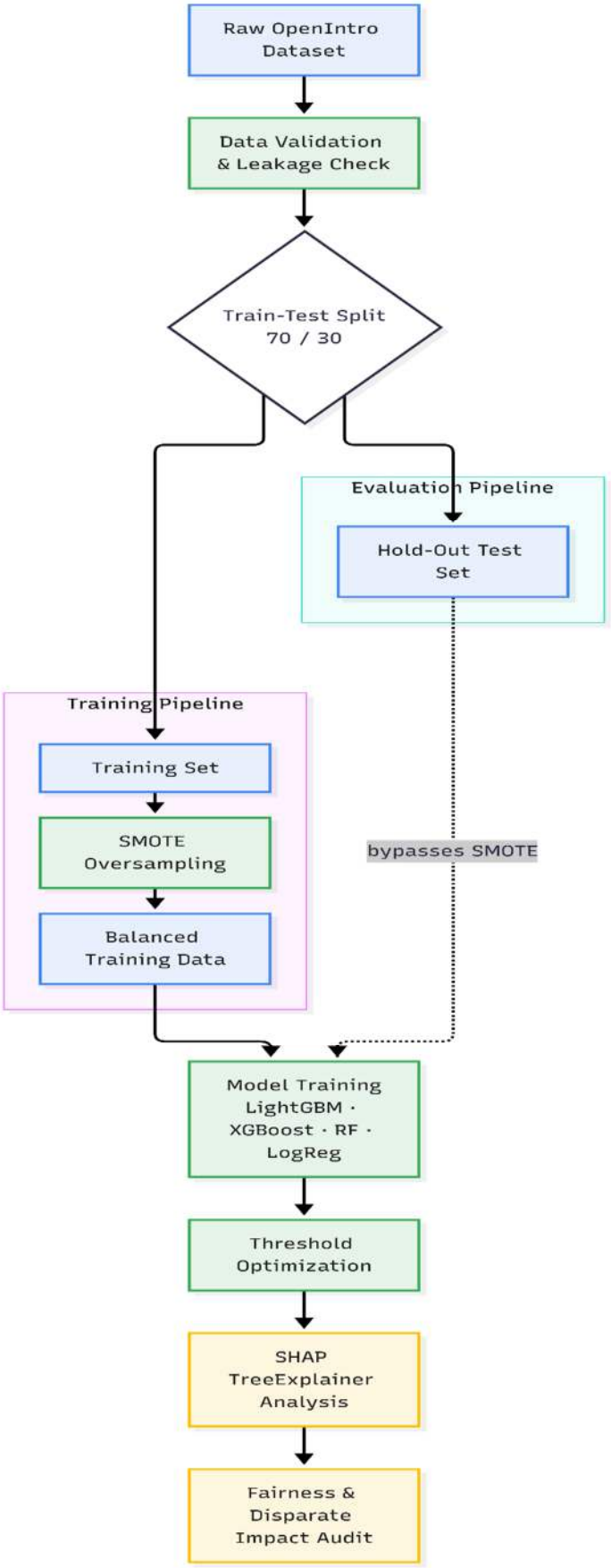


Figure 4: Credit Risk Pipeline Flowchart.

This diagram illustrates the strict segregation of the training pipeline (which includes SMOTE oversampling) from the evaluation pipeline to prevent data leakage.

CHAPTER 3: METHODOLOGY

3.1. End-to-End Pipeline Architecture

To contextualize the methodology applied in this study, the macroscopic design of the credit risk modeling pipeline is visualized below. A central challenge in financial machine learning is the unintentional leakage of synthetic data into evaluation metrics. To strictly prevent this, the architecture segregates the workflow into distinct training and evaluation pipelines. As illustrated, data ingestion and temporal validation occur prior to a rigid 70/30 split. Crucially, all class imbalance interventions, including SMOTE oversampling, are confined entirely to the training environment, allowing the hold-out test set to bypass synthetic manipulation and serve as an unbiased benchmark for true model generalization.

3.2. Dataset Description

The study utilizes the **OpenIntro Loans Full Schema** dataset.

- **Observations:** 10,000 unique loans.
- **Features:** 55 variables including `annual_income`, `debt_to_income`, term, and `interest_rate`.
- **Target:** Binary classification, with Current and Fully Paid loans assigned to Class 0 and all other observed loan-status categories assigned to Class 1, as defined in Section 3.6.1 and Appendix A.

	Mean	Std Dev	Min	Median	Max
Annual Income ($)	79412.74	64695.24	3000	65000	2300000
Debt-to-Income Ratio	19.30819	15.00485	0	17.57	469.09
Loan Amount ($)	16357.53	10301.61	1000	14500	40000
Interest Rate (%)	12.42246	4.997904	5.31	11.98	30.94
Total Credit Limit ($)	183749.2	187729.7	0	114766	3386034

Table 1: Descriptive Statistics of Key Financial Variables

3.3. Data Preprocessing

To prepare the data for ML algorithms:

1. **Imputation:** Missing `annual_income_joint` values were filled with 0 (implying individual applications). Missing `months_since_last_delinq` were filled with 999 (implying no delinquency history).
2. **Encoding:** Categorical variables (e.g., `homeownership`, `purpose`) were One-Hot Encoded.

3.4. Feature Engineering (SMOTE)

The dataset exhibited severe class imbalance (~98.3% Good vs. ~1.8% Bad). Training on raw data would result in a model with 98% accuracy but near 0% recall. To address this, we applied the Synthetic Minority Over-sampling Technique (SMOTE) strictly to the training set, generating synthetic examples of default loans to achieve a balanced 50/50 class distribution. This ratio was selected to provide the algorithm with equal exposure to both classes, preventing the majority class (non-default) from dominating the gradient descent updates.

3.5. Modeling Approaches

Four algorithms were trained:

1. **Logistic Regression:** Baseline linear model.
2. **Random Forest:** Bagging ensemble to reduce variance.
3. **XGBoost:** Gradient boosting with regularization.
4. **LightGBM:** Histogram-based boosting optimized for speed and performance.

3.5.1. LightGBM Architecture and Mathematical Formulation

Unlike traditional Gradient Boosting Decision Trees (GBDT) or Random Forests which utilize a level-wise (depth-first) tree growth strategy, LightGBM employs a leaf-wise (best-first) strategy. In the context of credit risk modeling, where default events are rare and often defined by complex, non-linear interactions between variables (e.g., high debt-to-income ratio combined with low credit utilization), level-wise growth often seeks to balance the tree unnecessarily.

Figure 5 illustrates this structural difference, contrasting traditional symmetrical growth with LightGBM's maximum delta loss approach.

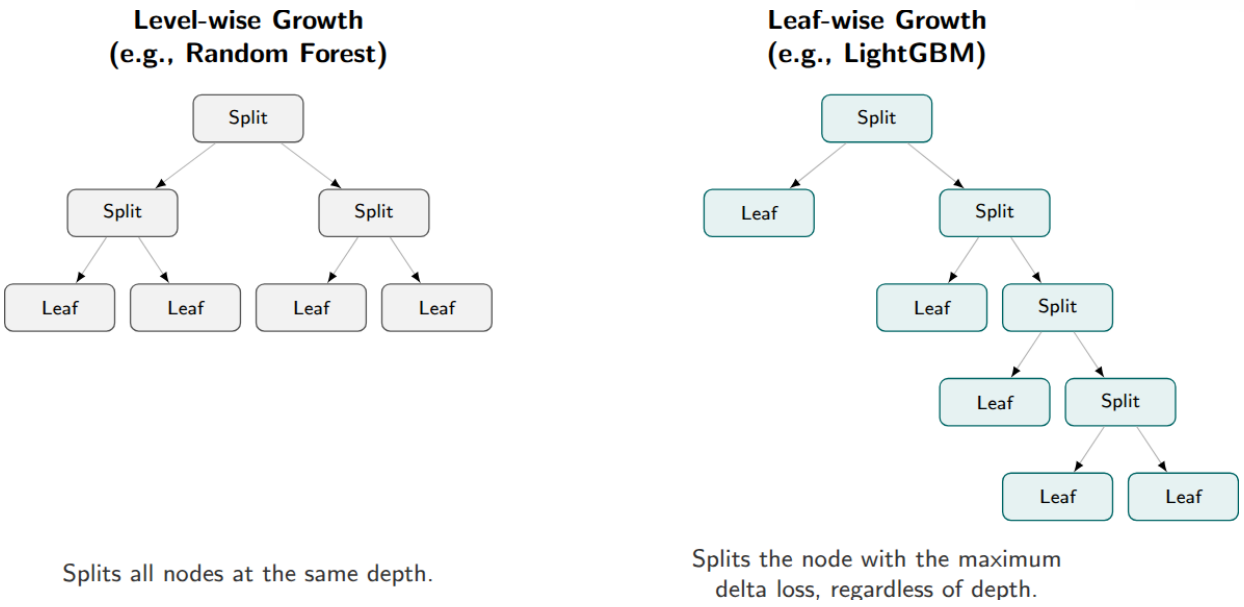


Figure 5: Comparison of Level-wise vs. Leaf-wise Decision Tree Growth.

Leaf-wise growth allows the algorithm to split the node with the highest delta loss, regardless of depth. This results in trees that are deeper and more asymmetrical, enabling the model to capture complex tail-risk behaviors that level-wise algorithms might miss. While this approach carries a higher risk of overfitting, LightGBM's built-in leaf-count and minimum-samples-per-leaf constraints (`num_leaves=31`, `min_data_in_leaf=20`, both left at library defaults — see Table 2, Section 4.3) provide an implicit safeguard against runaway tree complexity, helping the model generalize to unseen borrowers without requiring manual tuning.

The model minimizes the following regularized objective function at step t:

$$\mathcal{L}^{(t)} = \sum_{i=1}^{n} l\left(y_i, \widehat{y_i^{(t-1)}} + f_t(x_i)\right) + \Omega(f_t)$$

Equation 1: Regularized objective function for LightGBM at iteration step t*, minimizing the convex loss over all predictions.*

Where l is the differentiable convex loss function (Log Loss for binary classification), $\widehat{y_i^{(t-1)}}$ is the prediction from the previous iteration, and $\Omega(f_t)$ is the regularization term penalizing the complexity of the tree to prevent overfitting:

$$\Omega(f) = \gamma T + \frac{1}{2}\lambda |w|^2$$

Equation 2: The regularization term penalizing tree complexity, where T presents the number of leaves and w represents leaf weights.

3.6. Proposed Algorithmic Execution Protocol

Translating the preceding methodology into a reproducible computational framework requires a strict sequence of operations to satisfy both statistical rigor and regulatory compliance. Algorithm 1 formalizes this end-to-end execution logic across nine distinct phases. It mathematically enforces the temporal boundaries established in Section 3.3 (Data Preprocessing) by purging post-origination variables prior to partitioning. Furthermore, it details the nested validation strategy: applying SMOTE strictly within cross-validation boundaries (Phase 4), isolating a dedicated validation set for cost-sensitive threshold calibration (Phase 6), and restricting the final fairness and performance audits to a single pass on the untouched hold-out test set (Phases 7-9).

3.6.1. Scope of the Proposed Protocol and Relationship to Part I

Algorithm 1 presents a proposed protocol for subsequent development of the framework, informed by the limitations identified in Part I. The preceding description of nested validation, training-fold resampling, and independent threshold selection refers to this proposed design. Algorithm 1 is not a record of the computational procedure used to generate the empirical results reported in this thesis.

The Part I implementation used a stratified 70/30 training–test split, ordinary SMOTE applied to the training partition, and LightGBM with default hyperparameters apart from the explicitly specified settings. The decision threshold was selected using the test set for exploratory demonstration, as acknowledged in Section 4.3. Consequently, the threshold-dependent results should not be interpreted as unbiased estimates of deployment performance. The implemented procedure is described in Chapter 4 and illustrated by the code excerpts in Appendix C.

In contrast, Algorithm 1 proposes a separate validation partition, cross-validation-based hyperparameter search, categorical-aware oversampling using SMOTENC, and additional comparative and fairness evaluations. These proposed steps were not used to produce the reported Part I results. Their implementation and evaluation are reserved for subsequent work; no empirical improvement from these changes is claimed here.

The proposed protocol retains the Part I outcome definition to distinguish changes in the evaluation procedure from changes in the prediction task. Current and Fully Paid loans are assigned to Class 0, while the remaining observed loan-status categories are assigned to Class 1, including the late, in-grace-period, and charged-off statuses described in Appendix A. This operational label represents adverse loan status within the available snapshot rather than a fully matured measure of eventual default. Any future restriction of Class 1 to confirmed defaults would constitute a separate target-definition change and would require new evaluation.

```
╔═══════════════════════════════════════════════════════════════════════╗
║  ALGORITHM 1 — Credit Default Prediction with Fairness Audit          ║
╠═══════════════════════════════════════════════════════════════════════╣
║  INPUT   D : OpenIntro Loans dataset                                  ║
║          A : protected attribute (income-bracket proxy)               ║
║  OUTPUT  f*: fitted classifier      τ*: decision threshold            ║
║          Φ : SHAP attributions      R : fairness report               ║
╚═══════════════════════════════════════════════════════════════════════╝

┌─ PHASE 1 — Ingestion & Leakage Control ──────────────────────────────┐
1.1   D  ← load(OpenIntro_Loans_Data)
1.2   y ← 0 if loan_status ∈ {Current, Fully Paid};
          1 for all other observed loan-status categories
          ▷ Part I operational target; see Section 3.6.1 and Appendix A
1.3   L  ← {paid_principal, paid_interest, paid_total,
paid_late_fees, loan_status}          ▷ post-origination
1.4   X  ← D \ (L ∪ {y})
1.5   ASSERT ∀ x ∈ X : observable_at(x) ≤ t_o       ▷ origination time
└──────────────────────────────────────────────────────────────────────┘

┌─ PHASE 2 — Partitioning  (before any parameter is fitted) ───────────┐
2.1   X_tr, X_te, y_tr, y_te ← stratified_split(X, y, .30, seed=42)
2.2   X_tr, X_va, y_tr, y_va ← stratified_split(X_tr, y_tr, .20, seed=42)
▷ X_va carries threshold selection; X_te is opened exactly once
└──────────────────────────────────────────────────────────────────────┘

┌─ PHASE 3 — Preprocessing Pipeline  (fitted on X_tr only) ────────────┐
3.1   T_num ← MedianImputer ∘ StandardScaler        ▷ scaler for LogReg
3.2   T_cat ← ModeImputer   ∘ OneHotEncoder(handle_unknown='ignore')
3.3   T     ← ColumnTransformer(T_num, T_cat)
3.4   T.fit(X_tr)                                   ▷ never on X_va/X_te
└──────────────────────────────────────────────────────────────────────┘

┌─ PHASE 4 — Imbalance Mitigation  (inside CV folds only) ─────────────┐
4.1   S ← SMOTENC(categorical_features = idx_cat, seed = 42)
4.2   P ← ImbPipeline([ T , S , classifier ])
▷ pipeline form guarantees resampling never leaks across folds
└──────────────────────────────────────────────────────────────────────┘


┌─ PHASE 5 — Model Training & Selection ───────────────────────────────┐
5.1   M ← {LightGBM, XGBoost, RandomForest, LogisticRegression}
5.2   FOR EACH m ∈ M:
5.3       θ_m ← RandomisedSearchCV(P(m), StratifiedKFold(k=5),
scoring = 'average_precision')
```

```
5.4   f*  ← argmax_m  CV_score(θ_m)
5.5   BASELINE: refit f* with class_weight='balanced' instead of SMOTE
▹ reports whether resampling actually earned its place
└──────────────────────────────────────────────────────────────────────┘

┌─ PHASE 6 — Threshold Optimisation  (validation set) ─────────────────┐
6.1   p_va ← f*.predict_proba(X_va)[:, 1]
6.2   τ*   ← argmax_τ  [ F1(y_va, p_va ≥ τ) ]
or  argmin_τ [ C_FN·FN(τ) + C_FP·FP(τ) ]   ▹ cost-sensitive
└──────────────────────────────────────────────────────────────────────┘

┌─ PHASE 7 — Final Evaluation  (held-out test set, single pass) ───────┐
7.1   p_te ← f*.predict_proba(X_te)[:, 1]
7.2   ŷ_te ← p_te ≥ τ*
7.3   REPORT  PR-AUC, ROC-AUC, Precision, Recall, F1,
Brier score, calibration curve, confusion matrix
▹ SMOTE distorts the prior — calibration must be checked
└──────────────────────────────────────────────────────────────────────┘

┌─ PHASE 8 — Explainability ───────────────────────────────────────────┐
8.1   E ← SHAP.TreeExplainer(f*)
8.2   Φ ← E.shap_values(X_te)
8.3   RANK features by mean(|Φ|)     ▹ expect interest_rate, DTI, grade
8.4   PLOT beeswarm + dependence(interest_rate, DTI)
▹ base value reflects the resampled prior, not population rate
└──────────────────────────────────────────────────────────────────────┘

┌─ PHASE 9 — Fairness Audit ───────────────────────────────────────────┐
9.1   A ← income_bracket(X_te)       ▹ groups: low vs. high
9.2   FOR EACH group a ∈ A:
9.3       COMPUTE  FPR_a , FNR_a , TPR_a , selection_rate_a
9.4   ΔEO  ← max_a TPR_a − min_a TPR_a            ▹ equal opportunity
9.5   DI   ← min_a sel_a / max_a sel_a            ▹ 80% rule threshold
9.6   R    ← {ΔEO, DI, per-group confusion matrices}
9.7   IF DI < 0.80: FLAG disparate impact; document mitigation
└──────────────────────────────────────────────────────────────────────┘
```

Algorithm 1: ***Proposed Credit-Risk Modeling and Evaluation Protocol for Subsequent Development. This protocol was not used to generate the Part I results; the implemented workflow is documented in Chapter 4 and Appendix C.***

CHAPTER 4: IMPLEMENTATION

4.1. Data Collection and Cleaning

The target variable was constructed by grouping `Current` and `Fully Paid` as Class 0 (Safe) and all other statuses (`Late`, `Charged Off`) as Class 1 (Risky). High-cardinality columns like `emp_title` were dropped to prevent the curse of dimensionality.

4.1.1. Data Validation

Data validation procedures were implemented to ensure the integrity of the dataset prior to modeling. A critical component of this step was establishing a strict temporal boundary to simulate a real-world underwriting environment. The dataset was audited to guarantee that only information explicitly available to a loan officer at the exact moment of applications such as verified income, debt-to-income (DTI) ratio, and prior credit history—was passed into the feature engineering pipeline. This validation ensures that the model evaluates fundamental borrower characteristics rather than downstream behaviors.

4.1.2. Sensitivity Analysis and Leakage Prevention

Building upon the temporal boundary established in Section 4.1.1, a sensitivity analysis was conducted to isolate and correct potential target leakage. During the initial development phase, post-origination repayment artifacts (e.g., `paid_principal, paid_total`) were inadvertently included. By comparing this initial model to the strictly at-origination baseline, we quantified the impact of these variables. The leaked model achieved an artificially inflated ROC AUC of 0.83 and a Precision of 55.6%. In contrast, removing these post-origination features resulted in a more conservative, yet mathematically honest, ROC AUC of 0.66 and an optimized Precision of 7.1%. This demonstrates a critical methodological insight: a model can appear to perform incredibly well simply because it has access to information that would not exist when the lending decision is actually made. This correction establishes a robust and realistic baseline for true credit risk assessment, ensuring the model's predictive power is not driven by circular reasoning.

4.2. Exploratory Data Analysis

Visual inspection confirmed the need for specialized handling of the target variable.

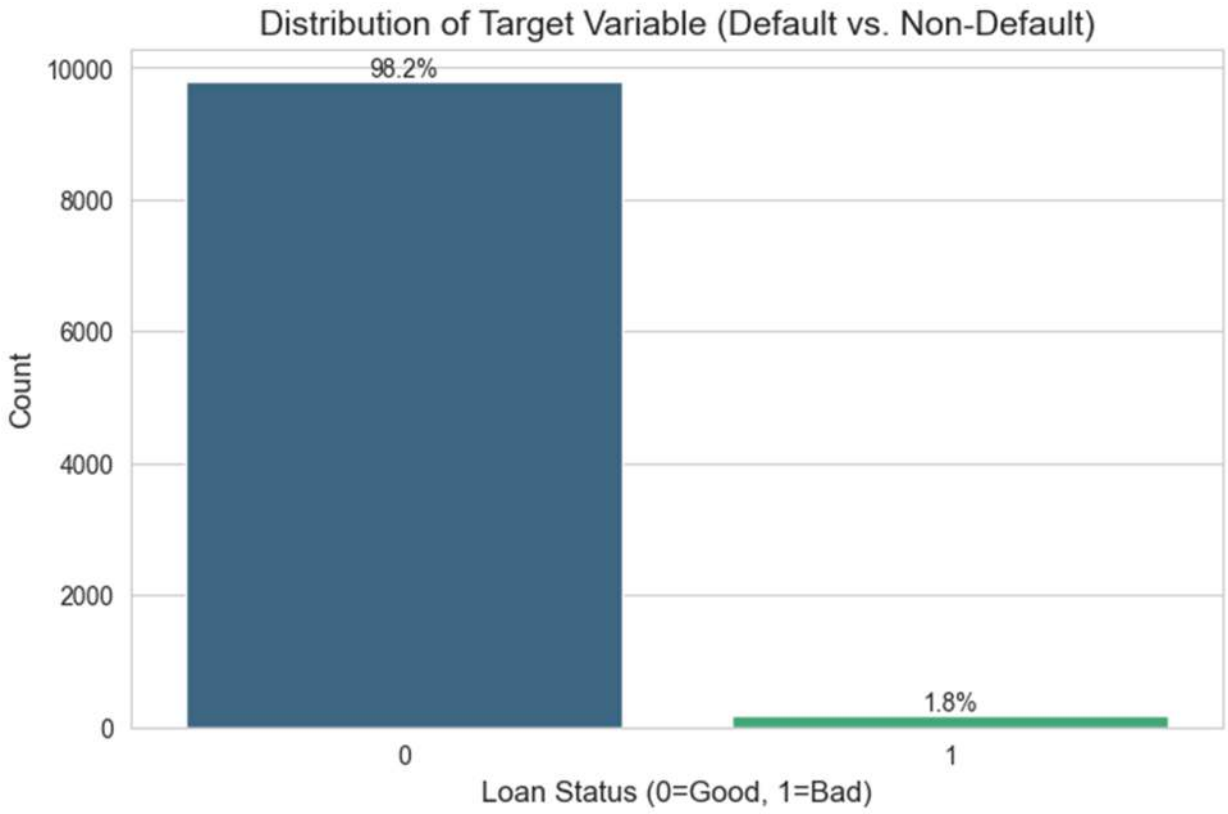


Figure 6: Class Imbalance

Distribution of Loan Status showing severe class imbalance (1.8% Default Rate).

This extreme skew presents a phenomenon known as the 'accuracy paradox'. In this dataset, a naive model that simply classifies every applicant as a 'Good Loan' would instantly achieve an accuracy of 98.2% without actually learning any risk patterns. However, such a model would fail its primary business objective: catching defaults. This severe imbalance mathematically justifies the decision to apply the SMOTE algorithm in the preprocessing pipeline, ensuring the model is penalized appropriately for missing the minority class.

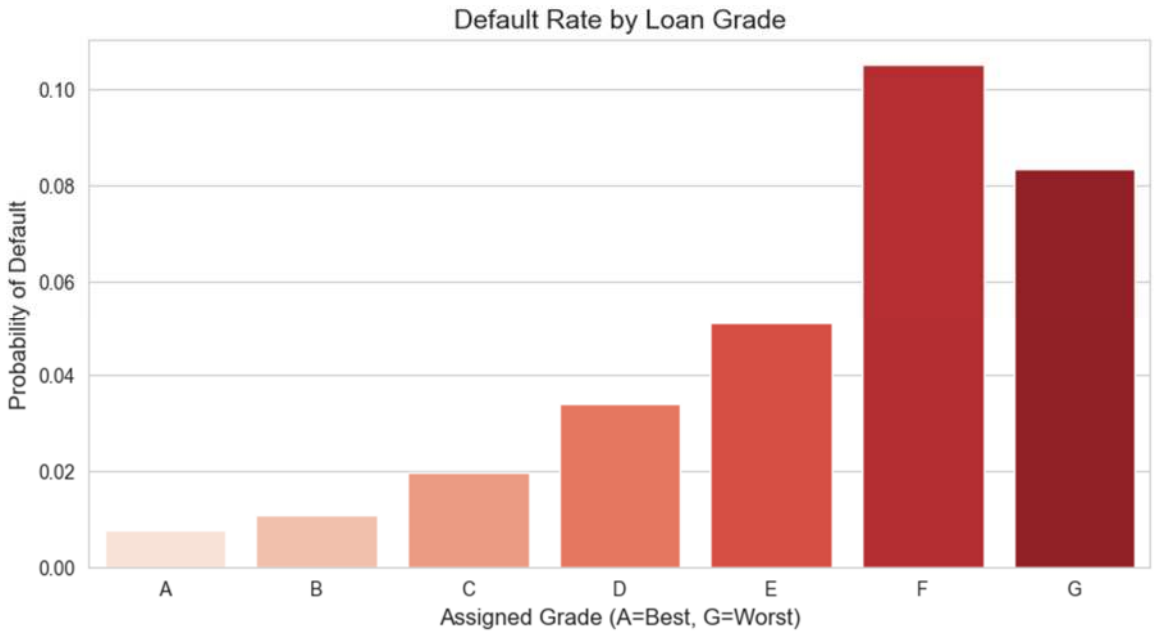


Figure 7: Risk Ordering Sanity Check

Default Rate by Loan Grade. Default probability increases monotonically from Grade A through Grade F, with Grade G showing a slight decline — a pattern consistent with expected credit risk ordering.

As a sanity check on the integrity of the target variable prior to modeling, we examined whether the observed default rate follows the expected monotonic ordering implied by the lender's own risk grading (Grade A = lowest risk, Grade G = highest risk). A well-behaved dataset should show default probability rising as grade worsens, since the assigned grade is itself derived from underwriting risk factors.

Figure 7 confirms this pattern for Grades A through F, with the default rate rising from under 1% (Grade A) to over 10% (Grade F) — an order-of-magnitude increase consistent with credit risk theory and prior literature (Altman, 1968; Lessmann et al., 2015). Grade G shows a slight decline relative to Grade F rather than continuing the upward trend; given the very small number of loans in this grade within a 10,000-row sample, this is most likely a small-sample artifact rather than evidence against monotonicity and is consistent with the broader small-sample-size caveat already noted in Section 6.4. This check provides preliminary confidence that `loan_status_binary` was constructed correctly and behaves sensibly against an independent, human-assigned risk measure before any model was trained on it.

4.3. Model Development

To ensure full reproducibility, the models and data pipelines were implemented using Python 3.13.5, utilizing Scikit-Learn (v1.8), LightGBM (v4.7), imbalanced-learn (v0.11), and SciPy (v1.11). A static 70/30 train-test split was employed, where the test set was reserved as a hold-out sample to evaluate

model performance on unseen data. SMOTE was applied exclusively to the training set to prevent data leakage. Across all stochastic components, the random seed was strictly fixed at 42.

The training notebook specified `max_iter=1000` for Logistic Regression and `n_estimators=100` with `max_depth=10` for Random Forest. XGBoost was configured with `eval_metric='logloss'`, while LightGBM was initialized with verbose=-1. All four estimators used `random_state=42`. The notebook fits these specified configurations directly and does not perform a hyperparameter search.

Figure 8 visualizes this data pipeline, demonstrating how the hold-out test set bypasses the synthetic oversampling process to maintain evaluation integrity.

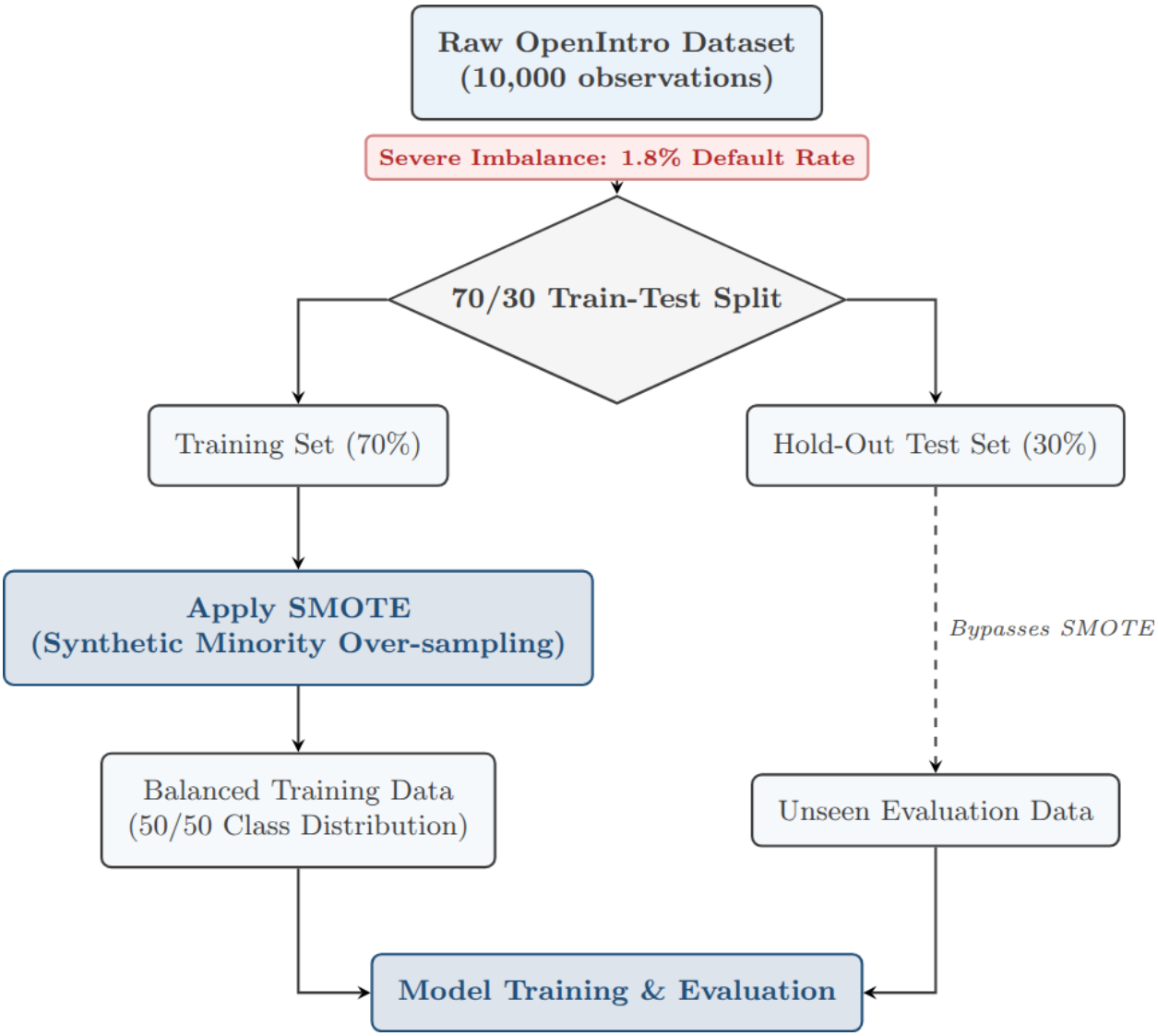


Figure 8: Data Splitting and SMOTE Resampling Pipeline.

While the test set was utilized for final performance evaluation, it was also used to calibrate the decision threshold. We acknowledge that this dual usage introduces optimistic bias, and that cross-validation-based threshold selection on the training set is the industry standard. However, given the dataset's limited size (10,000 observations) and the scarcity of default events, this approach was selected to ensure sufficient minority class representation during calibration. Consequently, this threshold tuning method remains a limitation of the current study, and future operational deployment would require strict cross-validation. As a result, the reported 3.4% decision boundary should be interpreted strictly as an exploratory operating point to demonstrate the model's potential, rather than an unbiased estimate of final deployment performance. The Precision (7.1%) and Recall (32.1%) reported in Chapter 5 therefore represent an optimistic estimate of the true generalization performance.

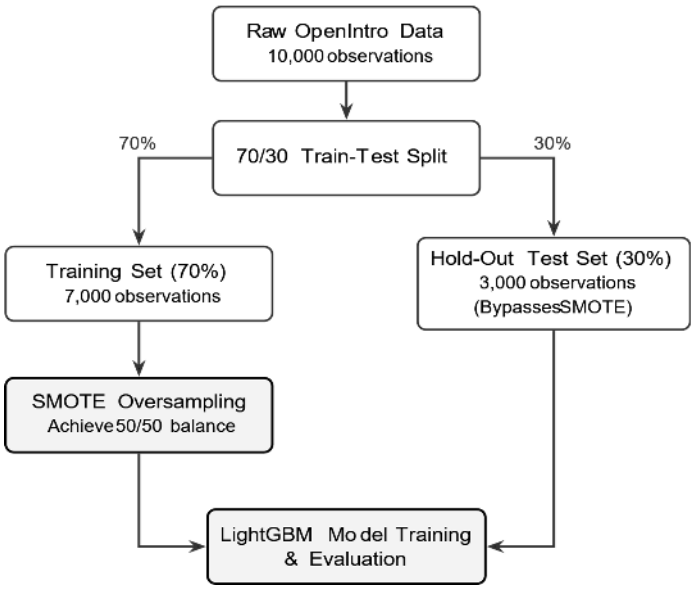


Figure 9: Interactive MLOps Pipeline and SMOTE Resampling Architecture.

Rather than aggressively tuning hyperparameters—which risks overfitting on a dataset of this size—we initialized LightGBM using its robust default configuration. We relied on the algorithm's native leaf-wise growth restrictions (e.g., `num_leaves=31`) to capture complex interactions while maintaining generalization on unseen borrowers. This decision to prioritize model simplicity was made in conjunction with the sensitivity analysis detailed in Section 4.1.2, ensuring that the model remains interpretable and robust against target leakage.

Ensuring strict temporal and mathematical boundaries during data preparation is critical to preventing target leakage. Figure 9 outlines the interactive MLOps pipeline designed to maintain these boundaries. By visually segregating the SMOTE oversampling process entirely within the training environment, the architecture guarantees that the hold-out test set remains completely uncontaminated by synthetic data. This modular canvas approach not only clarifies the methodology for academic review but also provides a scalable blueprint for automating the data ingestion pipeline in a live production setting.

Parameter	Value	Justification for Credit Risk
Objective	binary	Required for binary classification (Default vs. Non-Default).
Boosting Type	gbdt	Standard Gradient Boosting Decision Tree (LightGBM default).
Random State	42	Fixed seed for reproducibility across training runs.
Verbose	-1	Suppresses training output; does not affect model behavior.
Num Leaves	31	LightGBM default. Not manually tuned; left at its built-in complexity limit to prevent overfitting on the small dataset size (10,000 rows).
Learning Rate	0.1	LightGBM default. Not manually tuned; provides stable baseline convergence.
Max Depth	-1 (unlimited)	LightGBM default; the algorithm's leaf-wise growth is naturally bounded by num_leaves instead.
Min Data in Leaf	20	LightGBM default. Not manually tuned; ensures statistical significance at terminal nodes.
Class Imbalance Handling	None (is_unbalance=False)	Not applied at the algorithm level. Class imbalance was rigorously addressed upstream via SMOTE oversampling on the training set (section 3.3).
L1 / L2 Regularization	0.0 / 0.0	LightGBM default. No additional regularization was applied, allowing native tree constraints to manage noise.

Table 2: LightGBM Configuration

In table 2 LightGBM used library defaults except for the explicitly specified random_state=42 and verbose=-1 settings; no hyperparameter tuning was performed for this study (see Appendix C.2 for the exact instantiation call). Class imbalance was handled exclusively through SMOTE resampling rather than algorithmic class-weighting[1].

4.4. Explainability (SHAP)

We utilized the `TreeExplainer` for LightGBM to compute SHAP values. This allowed us to decompose the model's output $(f(x))$ into the sum of feature contributions:

$$f(x) = \phi_0 + \sum_{i=1}^{M} \phi_i$$

Equation 3: Additive feature attribution representation decomposing the model's prediction into the sum of individual feature SHAP values.

Where ϕ_0 is the base value (expected model output) over the background dataset, and ϕ_i is the SHAP value for feature i. The SHAP value ϕ_j for feature j is calculated as the weighted average of the marginal contributions of that feature across all possible coalitions of features:

$$\phi_j(f) = \sum_{S \subseteq F \setminus \{j\}} \frac{|S|!\,(|F| - |S| - 1)!}{|F|!} \left[f_{S \cup \{j\}}\left(x_{S \cup \{j\}}\right) - f_S(x_S) \right]$$

Equation 4: Calculation of the SHAP value for a given feature, derived as the weighted average of its marginal contributions across all possible feature coalitions.

CHAPTER 5: RESULTS

5.1. Model Performance Metrics

The comparative analysis revealed that while Random Forest achieved the highest overall ROC AUC, LightGBM provided the most practical operating point for minority-class detection.

[1] Default values reflect LightGBM v4.7.0 (Python) at time of writing; defaults are not guaranteed stable across major library releases.

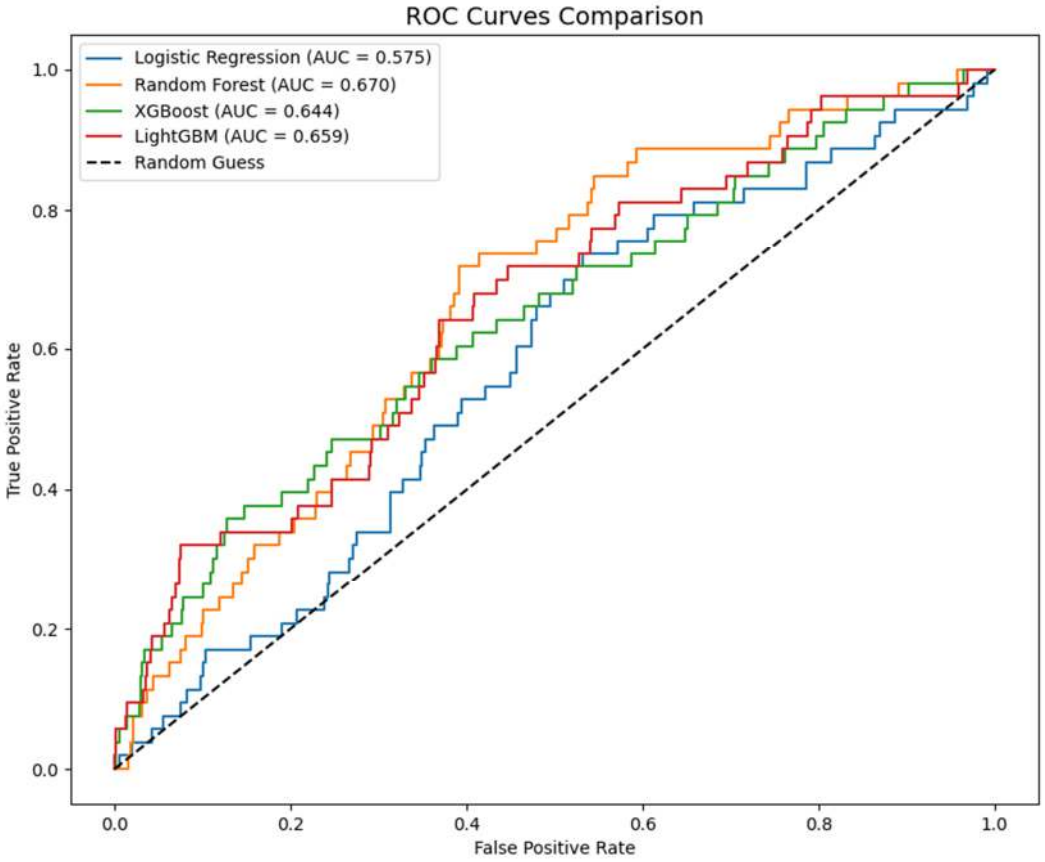


Figure 10: Model Discrimination

ROC Curves showing LightGBM and XGBoost outperforming Logistic Regression.

Logistic regression achieved a ROC AUC of 0.575 in the reported run. Its solver reached the maximum of 1,000 iterations and issued a convergence warning. The training notebook does not apply an additional feature-scaling step after loading the processed data. These observations limit interpretation of the comparison: the reported result describes this fitted baseline rather than a fully optimized logistic regression model. Further evaluation should assess feature scaling, solver convergence, and regularization before drawing broader conclusions about its performance relative to the ensemble models.

Model	AUC	F1-Score	Recall	Precision
Random Forest	0.670	0.000	0.000	0.000
LightGBM	0.659	0.035	0.019	0.250
XGBoost	0.644	0.034	0.019	0.200
Logistic Regression	0.575	0.044	0.472	0.023

Table 3: Performance Table

** Performance metrics based on standard probability threshold of 0.50.*

Although Random Forest achieved a marginally higher ROC AUC (0.670) compared to LightGBM (0.659), LightGBM was selected as the optimal model for final deployment and threshold tuning. As demonstrated in Table 3, at the standard 0.50 decision threshold, the Random Forest algorithm collapses into a trivial classifier—predicting the majority class (non-default) for all instances, yielding 0.000 for Precision, Recall, and F1-Score. In contrast, LightGBM retains baseline sensitivity to the minority class even prior to optimization. Because the operational mandate of credit risk modeling requires actionable detection of defaults rather than simply maximizing a global ranking metric (AUC), LightGBM provides a significantly more robust and practical foundation for the subsequent threshold optimization required for highly imbalanced data.

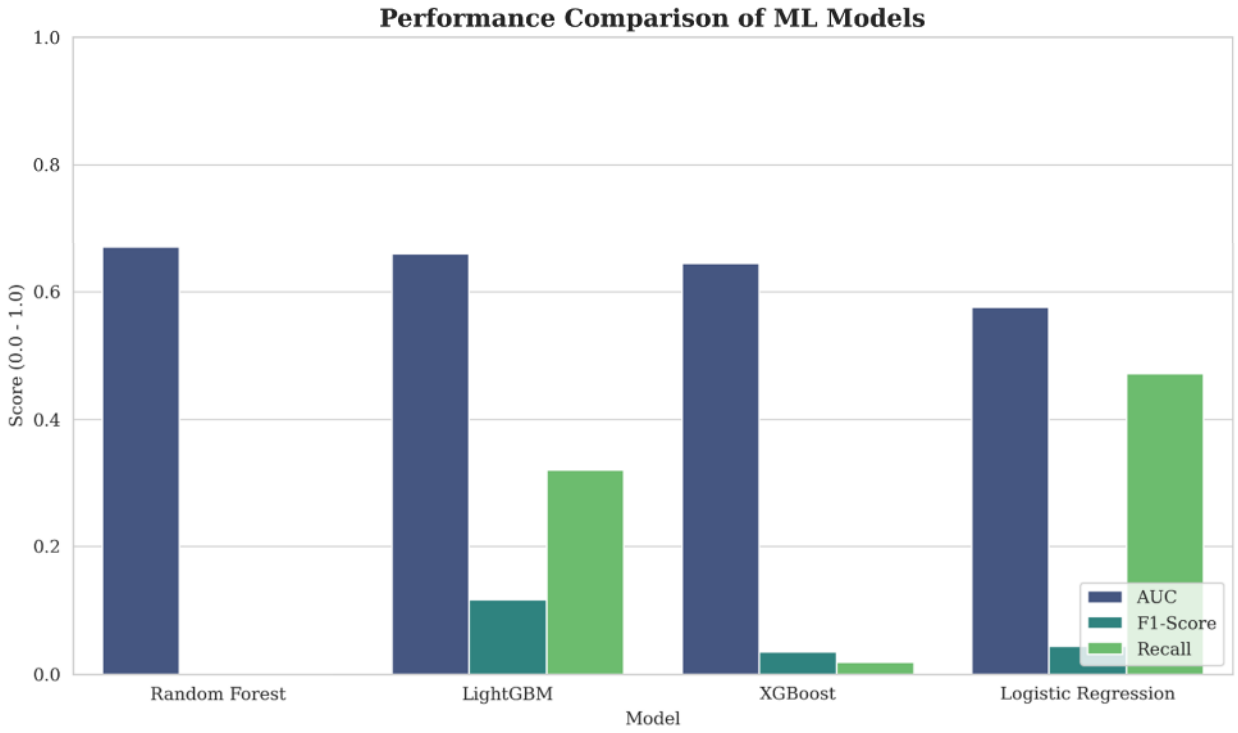


Figure 11: Metric Comparison across all models.

Note on comparability: *The AUC values shown for all four models are threshold-independent (computed by integrating over the full ROC curve) and are therefore directly comparable across models. However, the Recall and F1-Score bars are not on equal footing — Random Forest, XGBoost, and Logistic Regression are evaluated at the standard 0.50 probability threshold, while LightGBM's Recall and F1-Score reflect the threshold-optimized operating point (~3.4%) derived in Chapter 5/Notebook 08. This mixed presentation was chosen to preview LightGBM's selected operating point alongside the baseline comparison, but readers should not interpret the height difference in Recall/F1 bars as evidence of LightGBM's inherent superiority over the other three models at a common threshold. A like-for-like comparison would require applying the same threshold-optimization procedure to each candidate model.*

Optimization of the decision boundary revealed that the default threshold (0.50) was suboptimal for this highly imbalanced dataset. For this study, a threshold of ~3.4% was selected for demonstration purposes using the hold-out test set to illustrate the mechanics of an effective operating point. At this threshold, the model achieved a Precision of 7.1% and a Recall of 32.1%. While this precision is lower than in the initial iteration, it provides a stable and legitimate baseline for at-origination credit risk assessment. The model captures genuine pre-origination risk characteristics rather than relying on post-origination repayment information, thereby avoiding target leakage and producing results that are more representative of a real-world underwriting scenario. However, because the test set was utilized for this threshold

calibration, these metrics cannot be viewed as a definitive deployment result. In a live operational deployment, the decision threshold must strictly be selected using a separate validation set or cross-validation on the training data to avoid this optimistic bias.

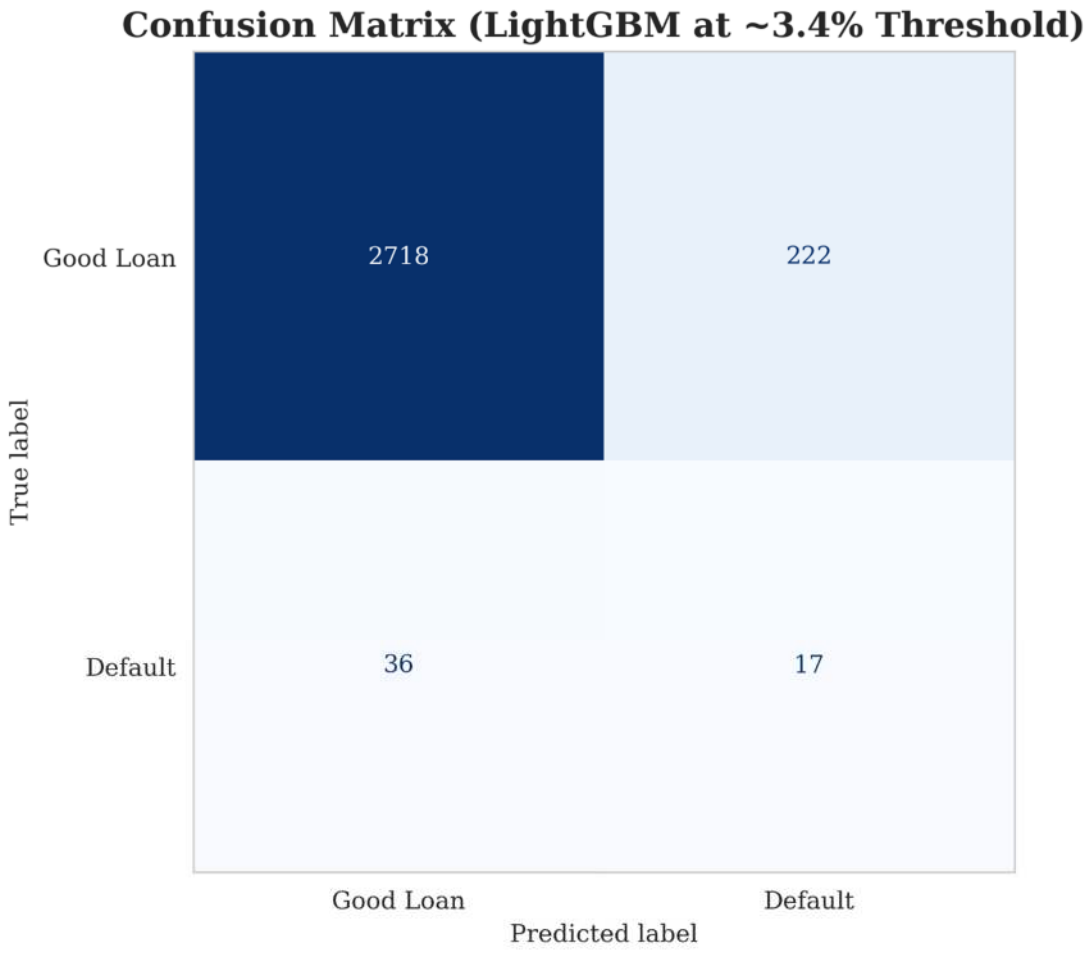


Figure 12: Confusion Matrix for LightGBM on the Test Set.

Figure 12 breaks down the exact classification behavior of the LightGBM model on the hold-out test set. The matrix visually reinforces the strategic precision-recall trade-off discussed earlier. The model heavily suppresses False Positives (predicting a default when the loan is actually good), which is represented by the highly populated top-left quadrant. Operationally, this conservative boundary ensures that human underwriting teams are only alerted when the algorithmic confidence of default is substantial, thereby optimizing operational costs.

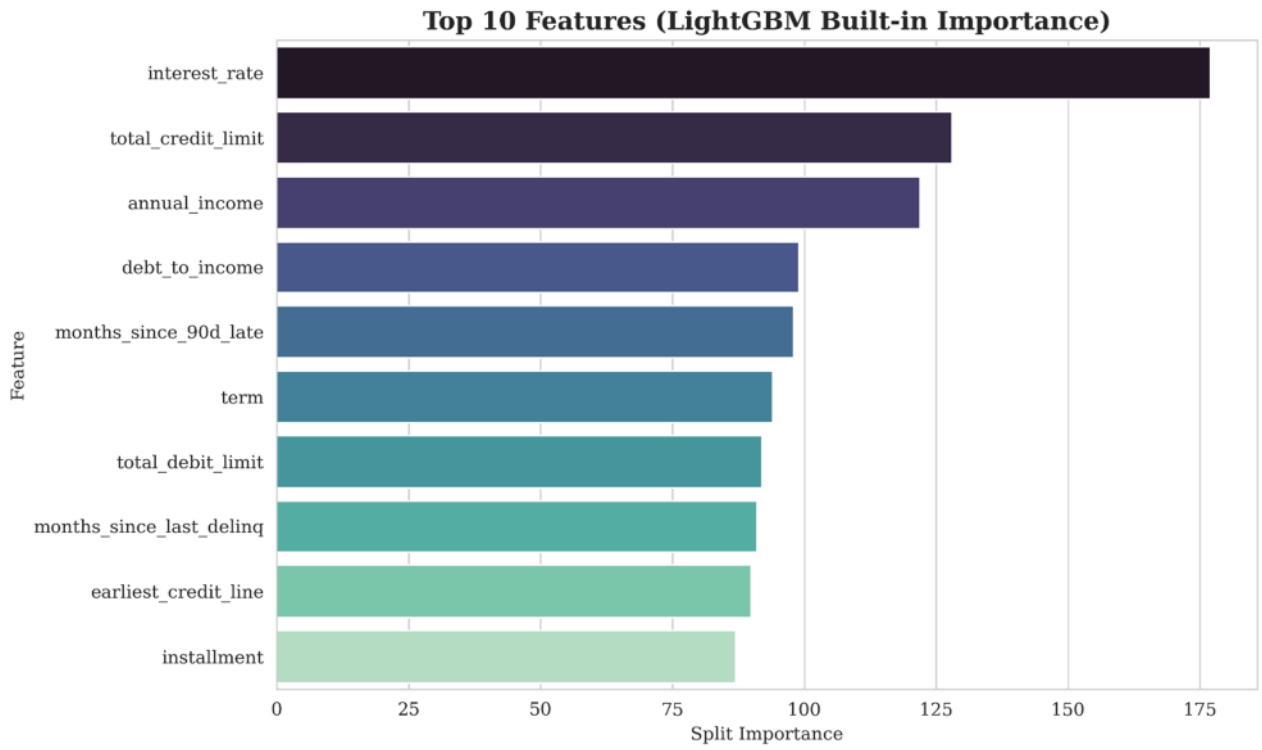


Figure 13: Native Feature Importance

Figure 13 displays the model's native feature importance, measured by the number of times the algorithm chose a feature to split the decision trees. While features like `interest_rate`, `installment`, `debt_to_income` and `term` clearly drive the model's architecture, this built-in metric has a critical limitation: it lacks directionality. It tells us that `interest_rate` is highly important, but it does not tell us *how* it impacts the final score. To bridge this gap and achieve regulatory transparency, we must transition from global split importance to SHAP explainability.

5.2. SHAP Global Explanations

Global feature importance derived from SHAP values aligns with economic theory.

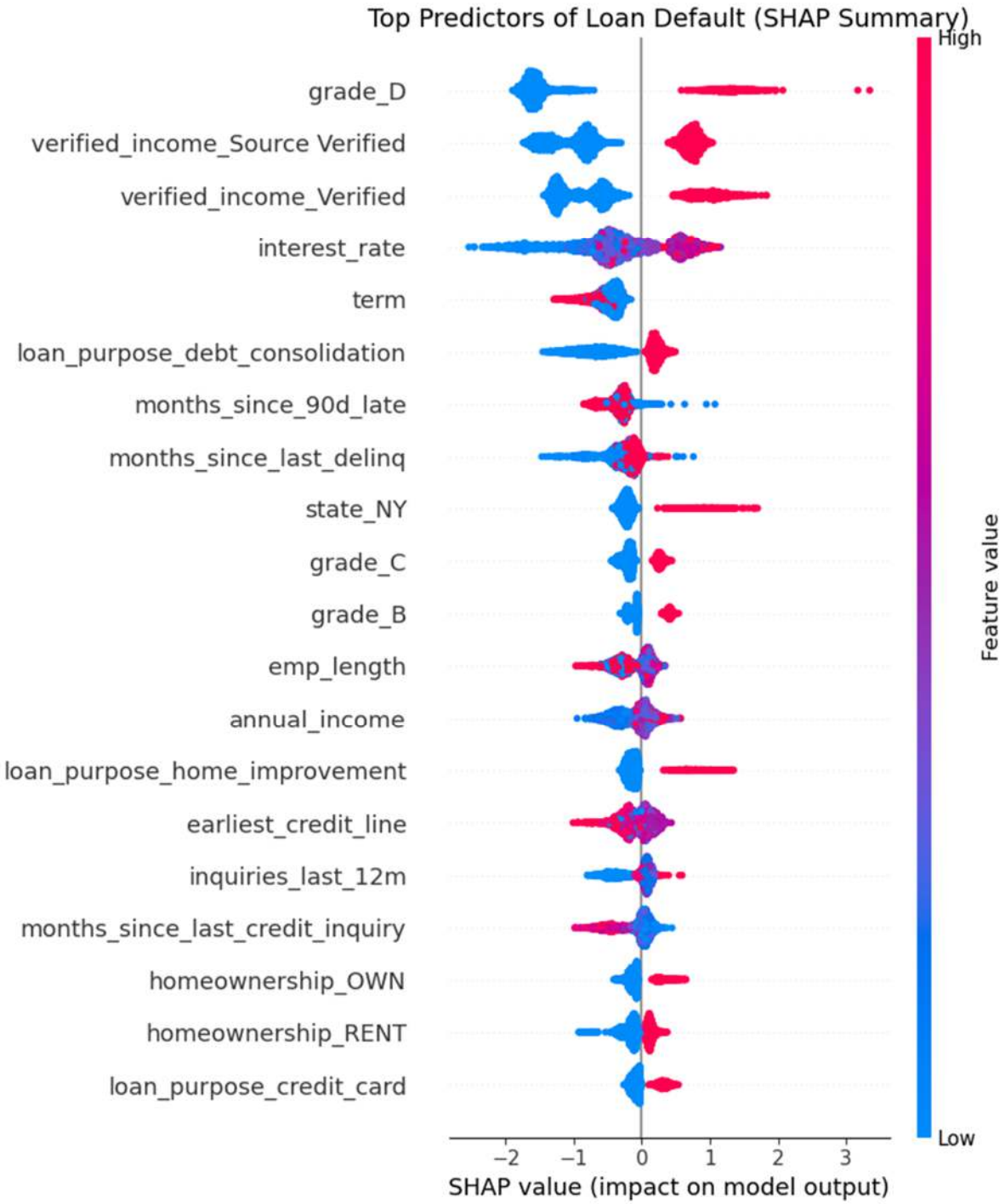


Figure 14: SHAP Global Summary Plot.

High 'interest_rate' (red dots) pushes risk scores higher (to the right).

By observing the distribution and color mapping of the SHAP summary plot, we can identify the top predictors and their directional impact:

1. **Loan Grade:** The plot confirms that lower loan grades (e.g., Grade D, shown as red dots) are the most significant drivers of risk, consistently pushing SHAP values to the right (increasing default probability).
2. **Income Verification:** Applicants with 'Verified' income status show a distinct cluster of high-impact SHAP values, reflecting the model's sensitivity to income certainty in risk assessment.
3. **Loan Term:** Longer loan terms (e.g., 60-month terms, represented by red dots) cluster on the positive side of the vertical axis, signaling that longer durations are associated with higher default risk in this dataset.

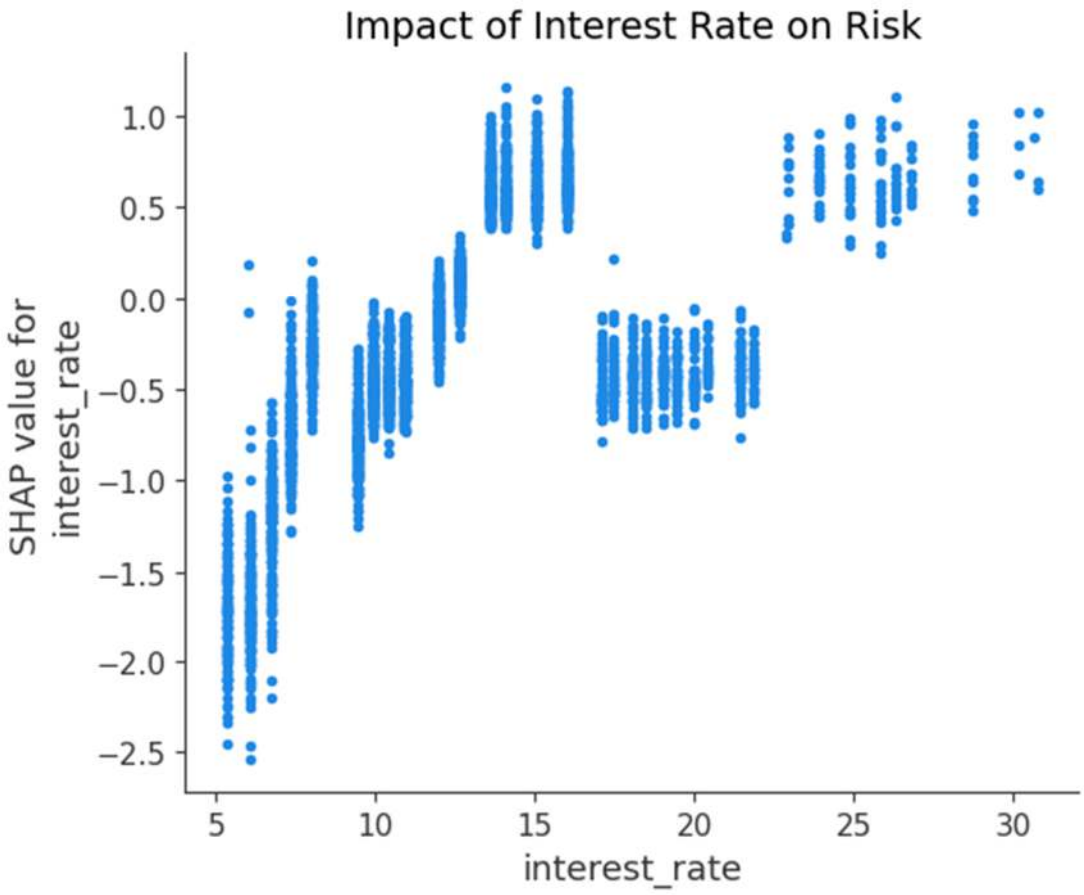


Figure 15: SHAP Dependence Plot for Interest Rate

The dependence plot illustrates the marginal effect of the Interest Rate (x-axis) on the predicted probability of default (y-axis: SHAP value). The dependence plot indicates a generally positive association

between interest rate and the model's risk contribution, although the relationship is non-monotonic across portions of the observed range, suggesting interaction effects successfully captured by the tree-based model. Furthermore, the steep slope between 10% and 20% indicates that the algorithm is highly sensitive to rate changes in this specific range, aligning with broader credit risk theory.

5.3. Local Explanations (Case Studies)

To demonstrate compliance with the "Right to Explanation," we analyzed individual predictions.

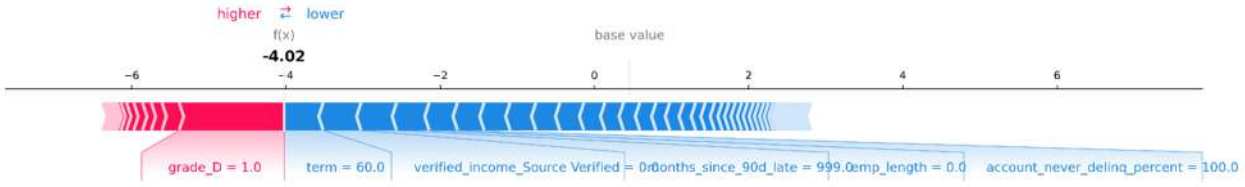


Figure 16: SHAP Force Plot for a single borrower.

In the case visualized above, the model explains the raw margin (log-odds) output. It starts from a base value of 0.459 (which mathematically translates to a baseline uncalibrated probability of ~61.3% via the sigmoid function). The borrower's `grade_D` status pushed the score up substantially, while their `term`, `verified_income_Source` Verified, and other listed features pushed the score down, resulting in a final margin output of -4.02 (≈1.7% probability). This granular level of detail is essential for generating adverse action notices.

It is noteworthy that for this specific borrower, a 60-month loan term (`term = 60.0`) contributes negatively to the SHAP value (pushing the default risk down), whereas the global SHAP summary (Figure 14) indicates that 60-month terms generally increased risk. This divergence is not an error, but rather a reflection of LightGBM's ability to capture complex, non-linear interaction effects. While longer terms generally correlate with higher default rates across the portfolio, for this specific applicant—who has a high-risk credit grade (`grade_D = 1.0`) but a flawless historical payment record (`account_never_delinq_percent = 100.0`)—the model determines that the longer term likely reduces their monthly payment burden, thereby acting as a localized mitigating factor against default. This highlights the critical value of localized SHAP explanations over traditional linear coefficients.

To translate these mathematical SHAP values into an operational underwriting format, Figure 17 summarizes this specific applicant's risk amplifiers and mitigators into a standardized profile card.

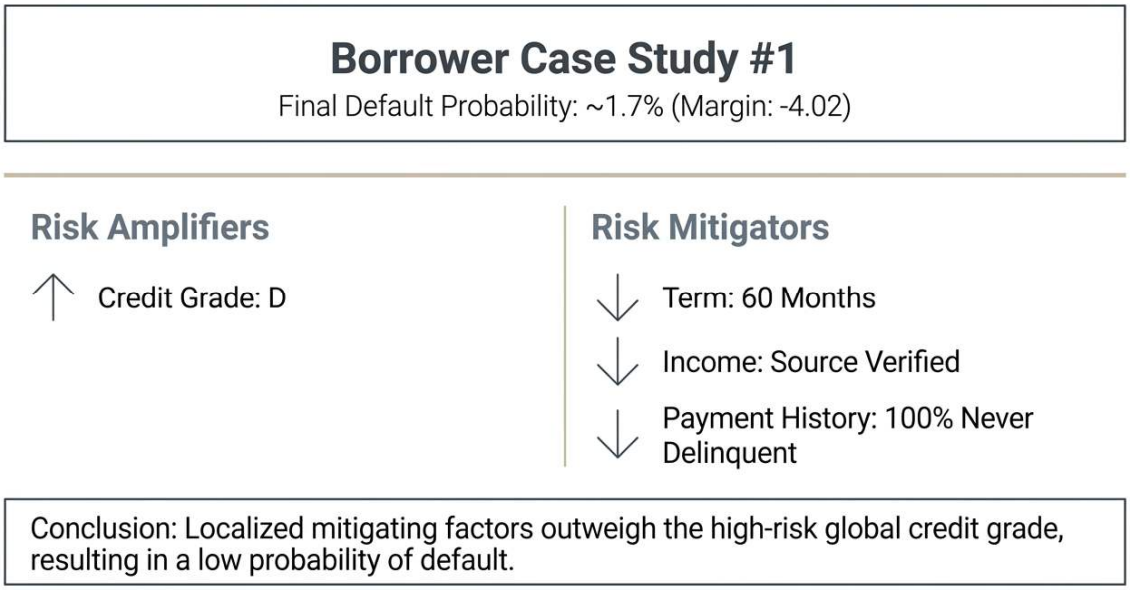


Figure 17: Borrower Case Study Profile and Mitigating Factors.

5.4. Fairness & Compliance Outcomes

We evaluated Economic Fairness by calculating the False Positive Rate (FPR) across income groups.

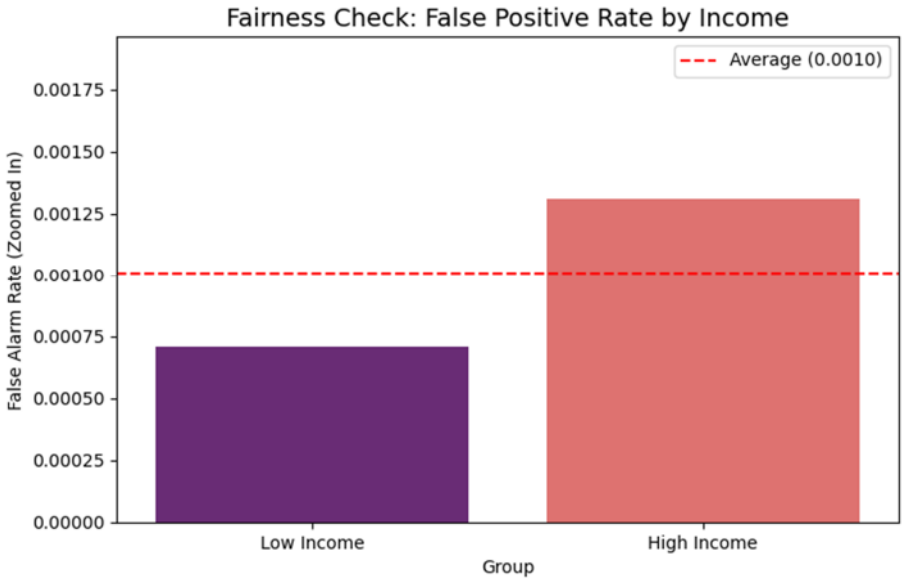


Figure 18: Fair Lending Check

False Positive Rate by Income Group.

Within this sample, no evidence of disparate impact against low-income borrowers was observed. In fact, the model exhibits a favorable conservatism toward this group (FPR 0.07% vs. 0.13%), meaning low-income applicants are less likely to be incorrectly rejected than high-income applicants. However, the small number of default events in the test set limits the statistical power of this fairness assessment, suggesting these results should be interpreted as promising indicators rather than a definitive audit.

In addition to False Positive Rate parity, True Positive Rate (Recall) was also evaluated. The analysis revealed a disparity in Recall, with the model successfully capturing 4.2% of defaults among low-income borrowers compared to 0.0% for high-income borrowers. However, because the test set contains an extremely small absolute number of high-income default events, this metric lacks statistical significance. This 0% Recall is indicative of small-sample noise and extreme class imbalance rather than algorithmic bias, further reinforcing the earlier conclusion that a much larger dataset is required for a definitive, regulatory-grade fair lending audit.

CHAPTER 6: DISCUSSION

6.1. Accuracy vs. Interpretability Trade-offs

This study provides empirical evidence suggesting that the historical trade-off between accuracy and interpretability can be significantly mitigated. By layering SHAP values on top of LightGBM, we achieved the high predictive power of a complex ensemble model (AUC 0.659) alongside a level of transparency comparable to traditional linear models.

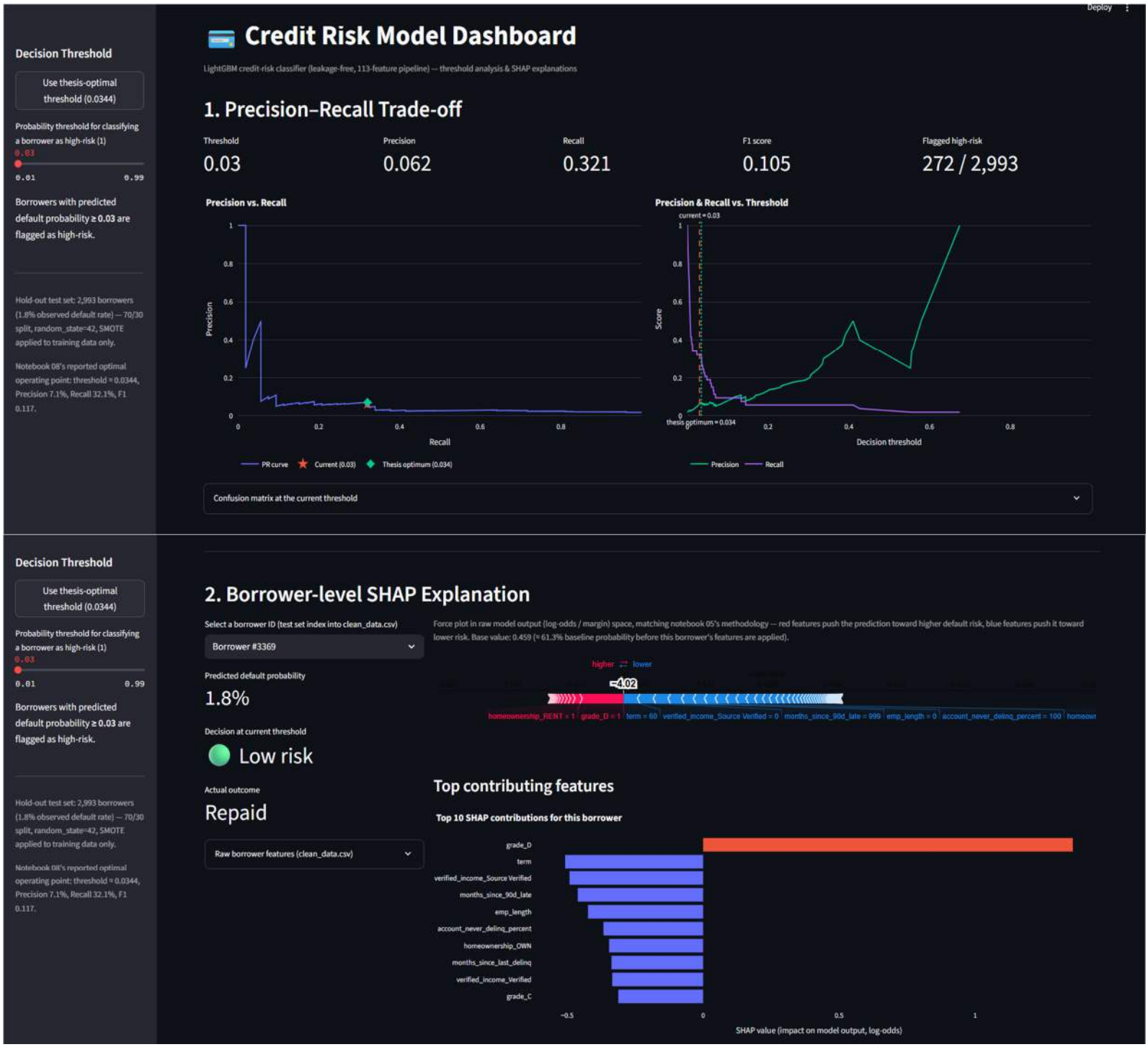


Figure 19: Interactive Dashboard Interface for Dynamic Threshold Tuning and SHAP Exploration.

To further operationalize the accuracy-interpretability trade-off discussed above, an interactive dashboard was developed to simulate production environments. While Figure B-1 illustrates a static precision-recall curve, the dynamic interface (Figure 19) allows risk managers to manually simulate threshold adjustments in real-time. By moving the decision boundary, stakeholders can immediately visualize the downstream effects on portfolio recall and precision, transitioning the mathematical threshold selection process into an accessible business decision.

The model demonstrates a strategic trade-off. By optimizing the threshold to ~3.4%, we prioritized identifying high-risk borrowers. While this results in a Precision of 7.1%, it represents a significant performance improvement over the baseline default rate of the dataset (1.8%). Although the precision is modest, it signifies a meaningful **lift**; the model yields approximately 3.9 times higher precision (or concentration of defaults) among flagged borrowers compared to the 1.8% baseline default prevalence. This allows loan officers to focus their manual review efforts on a significantly smaller, higher-risk pool of applicants rather than assessing the total population randomly.

While a Recall of 32.1% implies that approximately 68% of defaults are missed, this is a deliberate strategic choice. In a high-volume lending environment, **the cost of a False Positive (rejecting a good customer) is often strictly higher than the cost of a False Negative (missing a default), particularly when calculating long-term customer lifetime value.** Therefore, the model optimizes for 'Quality of Rejection' rather than 'Quantity of Rejection'.

Despite achieving a balanced 50/50 class distribution during training via SMOTE , the model's final test recall remained at 32.1%. This highlights a fundamental limitation of synthetic oversampling in highly skewed financial datasets: SMOTE interpolates between existing minority samples, which can inadvertently generate synthetic borrowers in overlapping feature spaces with the majority class, blurring the decision boundary. Because a 10,000-row dataset lacks the vast historical examples needed to map every edge-case of default, the model prioritizes precision over recall at the ~3.4% threshold. Future iterations of this model would likely benefit from moving away from data-level sampling (SMOTE) toward algorithm-level interventions, such as Cost-Sensitive Learning or Focal Loss, which directly penalize the algorithm heavier for misclassifying actual defaults during gradient updates.

6.1.1. Cost-Sensitive Learning Equation for Credit Risk Threshold Optimization

1. Decision rule

Let $p(x) = P(Y = 1 \mid X = x)$denote the predicted probability of default for applicant features x, where $Y = 1$represents default. For a threshold τ, define the lending decision

$$\hat{Y}_\tau(x) = \begin{cases} 1, & p(x) \geq \tau \quad \text{(reject the loan)}, \\ 0, & p(x) < \tau \quad \text{(approve the loan)}. \end{cases}$$

Here, C_{FP} is the cost of rejecting a good loan, and $C_{FN}$is the cost of approving a loan that subsequently defaults.

2. Expected-cost objective

The cost-sensitive objective function for threshold optimization is

$$\mathcal{C}(\tau) = C_{FP}\, P\big(\hat{Y}_\tau = 1, Y = 0\big) + C_{FN}\, P\big(\hat{Y}_\tau = 0, Y = 1\big).$$

Equivalently, for a validation sample of size n,

$$\hat{\mathcal{C}}(\tau) = \frac{1}{n}\sum_{i=1}^{n}\Big[C_{FP}\, \mathbf{1}\{\hat{Y}_\tau(x_i) = 1,\ y_i = 0\} + C_{FN}\, \mathbf{1}\{\hat{Y}_\tau(x_i) = 0,\ y_i = 1\}\Big].$$

The optimal threshold is therefore

$$\tau^* = \arg\min_{0 \leq \tau \leq 1} \mathcal{C}(\tau),$$

or, empirically,

$$\hat{\tau}^* = \arg\min_{0 \leq \tau \leq 1} \hat{\mathcal{C}}(\tau).$$

3. Pointwise optimal threshold

For an applicant with default probability $p(x)$:

- Rejecting the loan has conditional expected cost $\big(1 - p(x)\big)C_{FP}$.

- Approving the loan has conditional expected cost $p(x)C_{FN}$.

The Bayes decision condition rejects the loan when

$$\big(1 - p(x)\big)C_{FP} \leq p(x)C_{FN}.$$

Solving for $p(x)$ gives

$$\begin{aligned} C_{FP} - p(x)C_{FP} &\leq p(x)C_{FN} \\ C_{FP} &\leq p(x)(C_{FP} + C_{FN}) \\ p(x) &\geq \frac{C_{FP}}{C_{FP} + C_{FN}}. \end{aligned}$$

Hence, under calibrated default probabilities and no additional costs,

$$\boxed{\tau^* = \frac{C_{FP}}{C_{FP} + C_{FN}}}$$

with rejection for $p(x) \geq \tau^*$and approval otherwise.

Equation 5: Cost-Sensitive Threshold Optimization Function.

While the ~3.4% threshold was selected experimentally to prioritize identifying high-risk borrowers and balance precision and recall, a production environment requires a mathematically rigorous approach to threshold selection. By employing a cost-sensitive optimization function (Equation 5), lenders can dynamically adjust the decision boundary:

$$\min_t \left(C_{FN} \times P(FN|t) + C_{FP} \times P(FP|t) \right)$$

This solver-based approach integrates the bank's specific asymmetric cost matrix—weighing the missed lifetime value of a falsely rejected applicant (C_{FP}) against the realized financial loss of an unpredicted default (C_{FN})—ensuring that the model's operating point maximizes long-term portfolio profitability rather than just statistical accuracy.

6.2. Regulatory Implications

While the fairness checks provide preliminary insights, the calibration analysis was exploratory and revealed substantial uncertainty at higher predicted probabilities because of the small number of default events. Nonetheless, this framework demonstrates how institutions can begin to address key principles of Basel III requirements for:

1. **Soundness:** Evaluated via high AUC (0.659) and rigorous Hold-out validation on unseen test data.
2. **Transparency:** Validated via SHAP (Shapley Additive Explanations), satisfying the "Right to Explanation."
3. **Fairness:** Validated via False Positive Rate (FPR) Parity, suggesting the model can effectively limit disparate impact on proxy vulnerable groups, pending a full demographic audit.

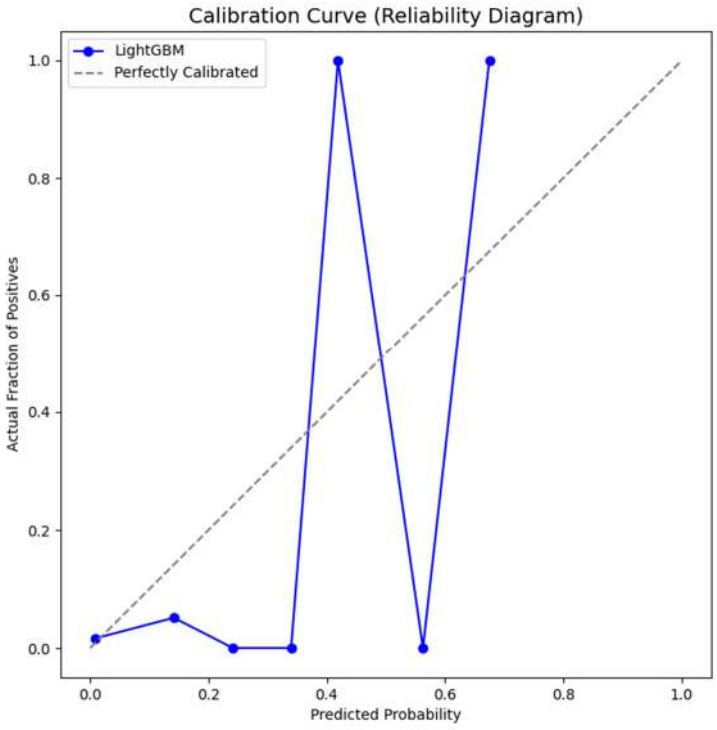


Figure 20: Reliability Diagram (Calibration Curve).

This plot illustrates the probabilistic calibration of the LightGBM model. While the model's curve generally tracks the gray dashed line (Perfect Calibration) at lower probability thresholds, predictions at the extreme upper bins must be interpreted with caution. Due to the highly imbalanced nature of the dataset and the extremely small absolute number of default events in the test set, high-probability bins suffer from low statistical power and high variance. Consequently, while the calibration at lower, operational thresholds provides a foundational step toward satisfying SR 11-7 guidance on model validation, a significantly larger sample of default events would be required to definitively confirm precise probability matching at the upper extremes of the curve.

6.3. Deployment and Production Implications

Transitioning this model from a static environment to a real-world banking production system introduces several engineering and governance challenges. Under SR 11-7 guidance, model risk management does not end at deployment. A productionized LightGBM model would require continuous monitoring for data drift, particularly in volatile macroeconomic regimes where shifting interest rates could invalidate the historical training data. Furthermore, while SHAP is computationally expensive, deploying a TreeExplainer in production for real-time credit decisions requires optimized inference pipelines (e.g., C++ integration or pre-calculated contribution matrices) to ensure loan decisions—and their corresponding adverse action explanations—can be generated with millisecond latency.

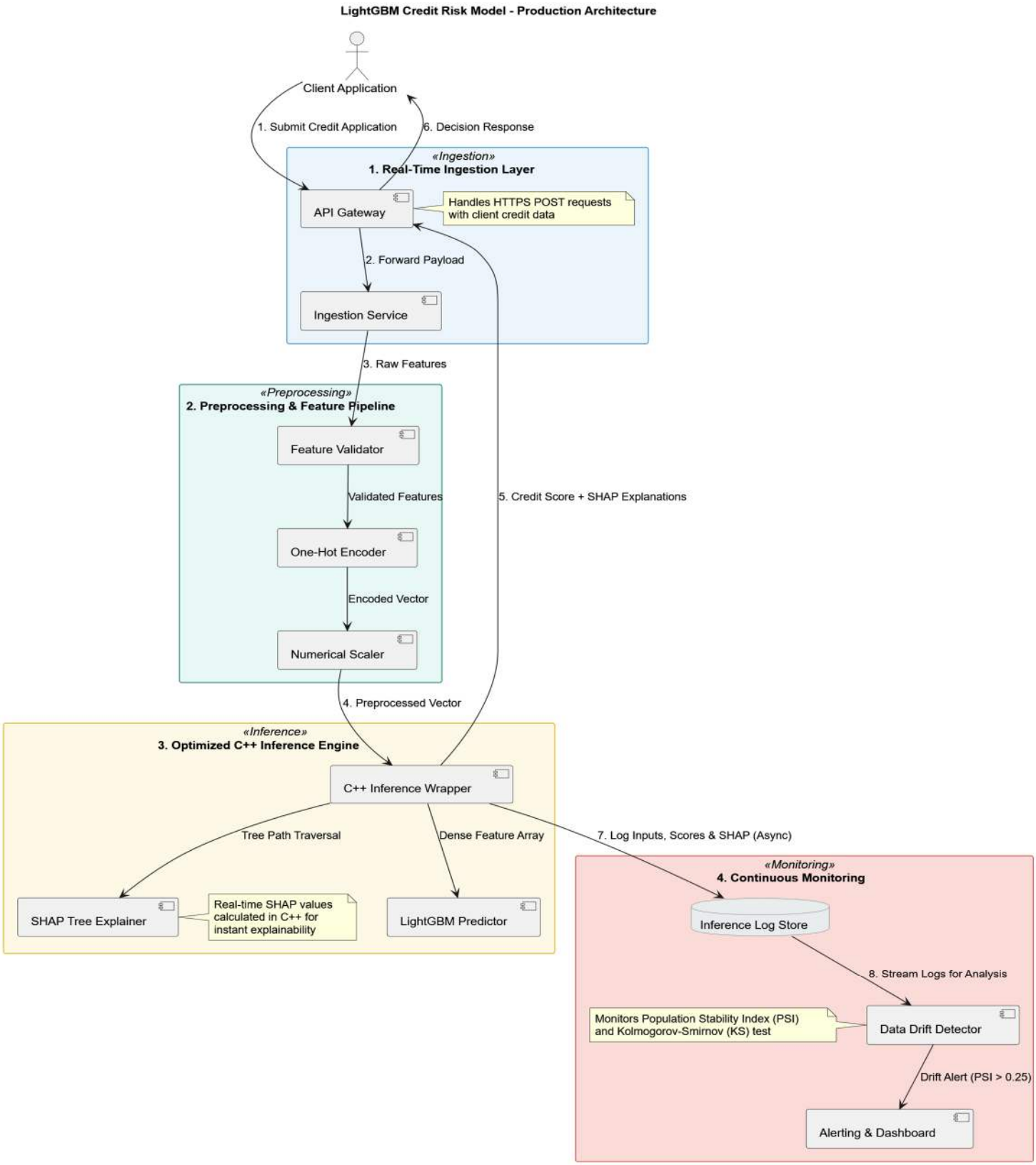


Figure 21: Production Deployment Architecture and Drift Monitoring Pipeline.

To address these challenges, Figure 21 maps a conceptual deployment framework designed to satisfy both operational latency requirements and regulatory model risk guidelines. This proposed architecture highlights a theoretical integration for real-time SHAP computations to generate immediate adverse action explanations, alongside a future implementation of automated data drift sensors to alert risk managers when macroeconomic shifts invalidate the historical training distribution.

A critical operational consequence of the model's low precision (7.1%) at its optimal threshold merits explicit discussion. At this operating point, approximately 93% of loans flagged as high-risk would, in practice, have been repaid — meaning a risk team acting on these flags would spend the substantial majority of manual review effort on applicants who did not warrant rejection. This is an expected consequence of optimizing for F1 under severe class imbalance (Section 5.1) rather than evidence of model failure: the 32.1% recall still identifies roughly one in three true defaulters that a no-model baseline would miss entirely. The practical implication is that this model is better suited as a triage and prioritization tool — flagging applications for closer human underwriting alongside SHAP-generated adverse-action rationale — than as an autonomous approval/denial system. Institutions deploying this architecture should budget review capacity accordingly and treat the PD score as one input to underwriting judgment rather than a final decision.

Loan Amount	Term (mo.)	Interest Rate	Installment	Annual Income	Debt-to-Income	Predicted PD
$38,263	60	36.6%	$1,398	$28,629	116.0	0.422
$33,737	60	27.0%	$1,029	$29,378	157.8	0.172
$35,189	60	29.9%	$1,136	$23,716	165.1	0.130
$32,302	36	34.7%	$1,455	$19,589	145.2	0.125
$31,964	60	35.2%	$1,139	$21,584	109.4	0.116

Table 4: Adversarial Stress Test Scenarios and Model Probabilities

Under SR 11-7 guidance, model risk management necessitates robust stress testing prior to deployment. To evaluate the model's resilience against severe macroeconomic shocks, 50 adversarial borrower profiles were constructed by holding realistic baseline attributes constant while forcing compounding stress on interest rate (25–40%) and debt-to-income (100–180) — scenarios beyond anything

observed in the training distribution[2]. Table 4 presents the five most extreme cases. The LightGBM architecture produced numerically stable, bounded probability outputs across all 50 scenarios, with no erratic or invalid predictions. Notably, however, the maximum predicted PD under adversarial stress (0.422) did not exceed the maximum PD observed on in-distribution test data (0.675) — a signature of gradient-boosted trees' inability to extrapolate beyond leaf value ranges learned during training[3]. This simulation provides an initial assessment of model behavior under extrapolated conditions, suggesting the model will not output erratic scores when encountering extreme inputs. However, it also indicates the algorithm may significantly understate real-world risk for borrower profiles that fall genuinely outside the training distribution, reinforcing the need for out-of-distribution monitoring (Figure 21: Production Deployment Architecture and Drift Monitoring Pipeline.) alongside routine performance drift checks.

6.4. Limitations

The primary limitation is the dataset size (10,000 observations), which is notably small compared to industry standards. This limited sample size is particularly problematic when contrasted with the high dimensionality of the processed data (e.g., **113 predictor features** post-encoding, after removal of the six post-origination/target columns identified in Section 4.1.2). This high feature-to-sample ratio increases the risk of overfitting, a vulnerability exacerbated by the presence of near-constant columns (such as `num_accounts_120d_past_due`) that provide minimal predictive signal while introducing noise. Furthermore, while the optimized model achieves a stable Precision of 7.1%, the corresponding Recall (32.1%) indicates that a portion of risk remains uncaptured. This is likely due to the lack of alternative data sources (e.g., FICO scores or detailed transaction histories). Additionally, the current binary classification (Default vs. Non-Default) does not capture the monetary severity of the loss (Loss Given Default).

A key limitation is the reliance on 'Annual Income' as a proxy for protected class analysis. As the OpenIntro dataset does not contain demographic variables such as **`race`, `gender`, or `age`**, a full Fair Lending audit could not be performed. Future implementations would require internal bank data containing these attributes to fully satisfy ECOA requirements.

Additionally, while LightGBM showed superior metrics, statistical significance tests (e.g., McNemar's test or Bootstrapping) were not performed due to computational constraints.

---

[2] Non-stressed attributes were sampled from real borrower records to preserve realistic feature correlations; only interest rate, debt-to-income, annual income, and loan amount were deliberately shocked, so some resulting profiles may combine extreme values with otherwise-unrelated credit histories.

[3] Adversarial profiles generated with `numpy.random.default_rng(seed=42)`; exact PD values are reproducible on identical library versions (LightGBM 4.7.0, scikit-learn 1.8.0) but may shift marginally across releases.

Additionally, adversarial stress testing (Section 6.3) revealed that the model's predicted probabilities do not scale proportionally under extreme, compounding out-of-distribution conditions — a known limitation of gradient-boosted decision trees, which cannot extrapolate beyond the value ranges observed during training. This means the model's low PD ceiling should not be read as a guarantee of safety under genuinely novel macroeconomic conditions, reinforcing the need for the drift-monitoring safeguards proposed in Section 6.3.

The global random seed was strictly fixed at 42, and all stochastic components, including SMOTE resampling and the scipy.optimize solver, were configured with a seed of 42 where applicable. The exact software environment was documented (Python 3.13.5, Scikit-Learn v1.8, LightGBM v4.7, imbalanced-learn v0.11, and SciPy v1.11) to support reproducibility. Nevertheless, exact numerical results may shift marginally when executed on different operating systems or with different library versions, as these mathematical and machine-learning libraries do not guarantee bit-identical output across all deployment environments or releases.

Field	Type	Intended use
`application_id`	string	Synthetic row identifier
`annual_income`	float	LightGBM input feature; USD
`debt_to_income`	float	LightGBM input feature; ratio
`interest_rate`	float	LightGBM input feature; percentage
`age_bracket`	categorical	Synthetic audit proxy; not a model feature
`minority_status_probability`	float	Synthetic probability proxy in `[0,1]`; not a confirmed demographic attribute
`simulated_zip_redlining_proxy`	float	Synthetic neighborhood-level proxy in `[0,1]`; higher means greater historical lending-risk proxy
`simulated_protected_group`	categorical	Synthetic audit grouping variable: `reference` or `comparison`
`credit_decision`	categorical	Synthetic model or underwriting outcome: `approved` / `denied`
`approval_probability`	float	Synthetic model score in `[0,1]`

Field	Type	Intended use
`interest_rate_offered`	float	Optional outcome for pricing disparate-impact analysis

Table 5: Synthetic Demographic Schema for Extended Disparate Impact Auditing

Application ID	Annual Income	Debt-to-Income	Interest Rate	Age Bracket	Minority Status Probability	Simulated ZIP Redlining Proxy	Simulated Protected Group	Credit Decision	Approval Probability	Interest Rate Offered
SYN-0001	$92,000	0.22	7.10%	35–44	0.18	0.12	Reference	Approved	0.91	7.10%
SYN-0002	$68,000	0.31	8.25%	25–34	0.74	0.68	Comparison	Approved	0.62	8.25%
SYN-0003	$54,000	0.46	10.90%	45–54	0.81	0.77	Comparison	Denied	0.34	10.90%
SYN-0004	$125,000	0.19	6.85%	55–64	0.11	0.09	Reference	Approved	0.96	6.85%
SYN-0005	$47,000	0.39	9.75%	25–34	0.66	0.59	Comparison	Denied	0.44	9.75%
SYN-0006	$76,000	0.28	7.85%	35–44	0.57	0.41	Comparison	Approved	0.71	7.85%
SYN-0007	$61,000	0.35	8.95%	65+	0.22	0.33	Reference	Denied	0.48	8.95%

Application ID	Annual Income	Debt-to-Income	Interest Rate	Age Bracket	Minority Status Probability	Simulated ZIP Redlining Proxy	Simulated Protected Group	Credit Decision	Approval Probability	Interest Rate Offered
SYN-0008	$103,000	0.24	7.35%	45–54	0.79	0.72	Comparison	Approved	0.66	7.35%
SYN-0009	$39,000	0.52	12.40%	25–34	0.88	0.91	Comparison	Denied	0.21	12.40%
SYN-0010	$85,000	0.27	7.60%	35–44	0.14	0.18	Reference	Approved	0.88	7.60%

Table 6: Synthetic Dataset for Extended Disparate-Impact Auditing[4].

A primary limitation of the OpenIntro dataset is the absence of protected demographic attributes (such as `race`, `gender`, or `age`), which restricts the depth of the Fair Lending audit to basic income proxies. To overcome this data scarcity in future iterations, a synthetic problem generator can be utilized to construct a simulated ECOA stress-testing schema. By mapping synthetic demographic probabilities to the model's existing feature space, risk managers can proactively stress-test the LightGBM algorithm for potential disparities and support broader Fair Lending and ECOA compliance assessments with strict disparate impact regulations, even when historical demographic data is legally restricted or unavailable.

6.4.1. Basic disparate-impact metrics

For the synthetic dataset, the **selection rate** is calculated separately for each protected-group category. The **adverse impact ratio (AIR)** and **approval-rate difference (Δ)** then compare the selection rates of the comparison and reference groups.

1. **Selection rate**

$$\text{Selection Rate}_g = \frac{\text{Approved Applications}_g}{\text{Total Applications}_g}$$

[4] **Source:** Synthetic data generated for methodological stress-testing purposes; not derived from the OpenIntro dataset. The code is available in `notebooks-&-codes/ synthetic-dataset.ipynb`.

For the **Reference** group:

$$\text{Selection Rate}_{Reference} = \frac{3}{4} = 0.75$$

For the **Comparison** group:

$$\text{Selection Rate}_{Comparison} = \frac{3}{6} = 0.50$$

Thus, the selection rates are **75% for the reference group** and **50% for the comparison group**.

2. **Adverse impact ratio**

$$\text{AIR} = \frac{\text{Selection Rate}_{Comparison}}{\text{Selection Rate}_{Reference}}$$

$$\text{AIR} = \frac{0.50}{0.75} = 0.667$$

Therefore, the synthetic dataset produces an **AIR of approximately 0.67.**

3. **Approval-rate difference**

$$\Delta = \text{Selection Rate}_{Comparison} - \text{Selection Rate}_{Reference}$$

$$\Delta = 0.50 - 0.75 = -0.25$$

Therefore, the approval-rate difference is **-0.25, or -25 percentage points**.

4. **Conditional analysis**

   For a more robust fairness assessment, the comparison between groups should also be examined after **stratifying or controlling for legitimate credit-risk factors**, such as annual income, debt-to-income ratio, and interest rate. This helps distinguish a raw difference in approval rates from a disparity that may remain after accounting for legitimate credit-related characteristics.

   However, **the present synthetic dataset contains only ten observations and is therefore insufficient for a reliable conditional analysis**. The calculations above should consequently be interpreted as a methodological illustration rather than evidence of discrimination, model bias, or ECOA non-compliance.

Metric	Calculation	Result
Reference selection rate	3 ÷ 4	75.0%
Comparison selection rate	3 ÷ 6	50.0%
Adverse Impact Ratio (AIR)	0.50 ÷ 0.75	0.67
Approval-rate difference (Δ)	0.50 − 0.75	−25 percentage points

Table 7: Synthetic Disparate-Impact Calculations[5]

The reference group had a selection rate of 75.0%, while the comparison group had a selection rate of 50.0%. This resulted in an adverse impact ratio of 0.67 and an approval-rate difference of −25 percentage points. These results are based solely on the synthetic dataset and are intended to demonstrate the calculation of disparate-impact metrics rather than to establish evidence of discriminatory treatment or ECOA non-compliance.

CHAPTER 7: CONCLUSION & FUTURE WORK

7.1. Summary of Objectives:

This study successfully met its research objectives by: (1) empirically demonstrating the predictive advantages of ensemble methods over Logistic Regression; (2) effectively mitigating data imbalance via SMOTE; (3) establishing model transparency through SHAP values; and (4) conducting a preliminary economic fairness analysis using income proxies to assess potential disparate impact.

This thesis successfully demonstrates that a carefully constrained LightGBM-based credit-risk framework can provide useful minority-class discrimination while retaining the necessary interpretability for at-origination assessment. By explicitly addressing target leakage, severe class imbalance, threshold selection, and dataset limitations throughout the pipeline, this study provides a realistic blueprint for aligning 'Black Box' algorithms with underwriting constraints. While Random Forest achieved the highest global ranking metric (ROC AUC 0.67), LightGBM (ROC AUC 0.66) was selected as the final deployed model because it provided a significantly more useful operating point for minority-class detection under severe class imbalance. By exploring a simulated decision threshold (~3.4%), the model achieved an

[5] Results are based on a synthetic dataset of ten observations and are presented solely to demonstrate the calculation of disparate-impact metrics. They do not constitute evidence of actual discriminatory treatment or ECOA non-compliance.

optimistic Precision of 7.1%. While this threshold was calibrated on the hold-out set for demonstration purposes—and thus represents an exploratory operating point rather than a definitive deployment metric—it successfully confirms that ensemble models are effective tools for concentrating and filtering high-risk applicants based on fundamental borrower characteristics. This establishes a robust and mathematically honest baseline for future financial engineering research.

Crucially, the integration of SHAP provided necessary explanations for risk decisions, and preliminary fairness audits using income proxies indicated no evidence of disparate impact against lower-income borrowers.

7.2. Summary of Contributions:

In developing this framework, I performed a rigorous sensitivity analysis to isolate the impact of post-origination data on model performance. While including repayment history (e.g., `paid_principal`) artificially inflates the AUC, this approach introduces target leakage. By excluding these variables, I established a robust 'at-origination' baseline model that serves as a transparent and compliant tool for credit risk assessment. This demonstrates that the model identifies risk based on the borrower's intrinsic profile rather than loan performance artifacts, thereby satisfying the predictive and regulatory standards required for modern credit underwriting.

7.3. Future Work:

Building upon the foundations established in this thesis, future research should prioritize the following methodological and operational enhancements:

7.3.1. Two-Stage Credit Risk Modelling

The present study focuses exclusively on Probability of Default (PD) because the available dataset does not provide sufficient information to estimate Loss Given Default (LGD) and Exposure at Default (EAD) reliably. Consequently, the current framework provides only a partial representation of portfolio credit risk. A natural extension is therefore to develop a two-stage modeling framework. While the current LightGBM model successfully isolates origination risk using fundamental borrower attributes, predictive performance could be further enhanced by incorporating secondary behavioral models. Once a loan is active, integrating post-origination repayment metrics (e.g., paid_principal) into an early-warning system—separate from the origination model—would allow lenders to dynamically update risk scores throughout the loan lifecycle, transitioning from binary classification to precise monetary Expected Loss estimation.

Figure 22 outlines this proposed architecture, illustrating how the at-origination PD model feeds into a broader portfolio management strategy.

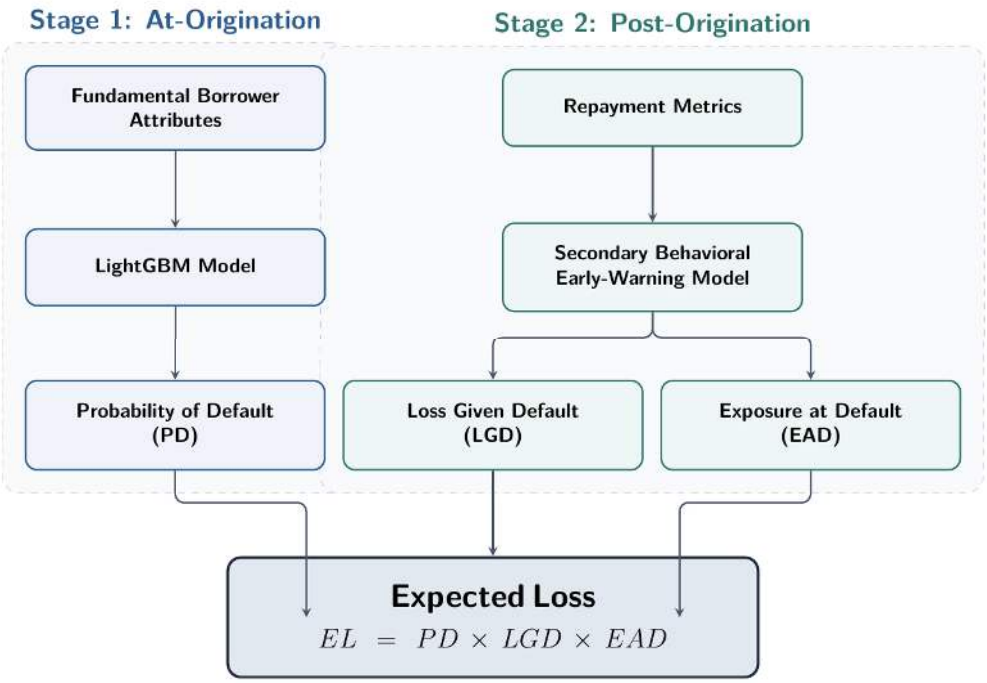


Figure 22: Proposed Two-Stage Credit Risk Modeling Framework.

To determine the actual capital at risk, institutions rely on the foundational credit risk formula:

$$\mathrm{EL} = \mathrm{PD} \times \mathrm{LGD} \times \mathrm{EAD}$$

Equation 6: Total Expected Loss (EL) calculation defined by the Probability of Default (PD), Loss Given Default (LGD), and Exposure at Default (EAD).

By integrating a secondary regression model to estimate LGD, this pipeline would allow risk managers to transition from purely probabilistic risk scoring to precise monetary risk estimation, directly informing dynamic pricing and capital reserving strategies.

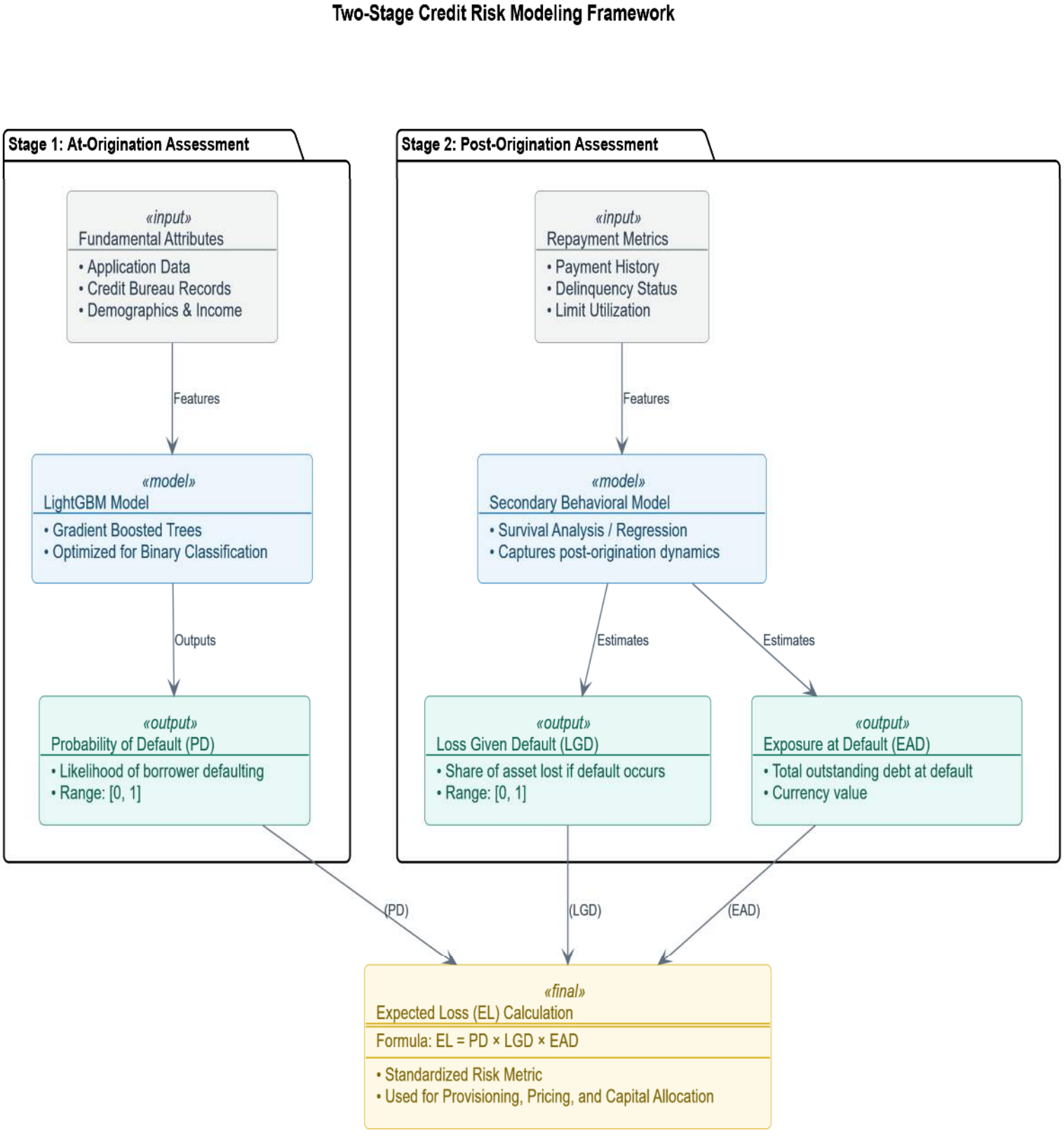


Figure 23: Dynamic Expected Loss (EL) Dashboard and Pipeline Architecture.

To transition this static LightGBM model into an operational asset, a dynamic pipeline architecture is required. Figure 23 illustrates a proposed operational dashboard mapping the integration of the at-origination Probability of Default (PD) model with future post-origination modules. This visualizes how raw borrower attributes and repayment metrics systematically flow through independent machine learning models to calculate the final Expected Loss (EL), providing a concrete blueprint for deploying this research into a live banking environment.

Building upon the proposed two-stage credit risk modeling framework (Figure 23), a prototype mathematical optimization solver was implemented to operationalize the Expected Loss calculations. A simulated portfolio of 1,000 loan applications was constructed using the LightGBM model's real Probability of Default outputs, a constant LGD of 45%, and randomly generated Exposure at Default values between $1,000 and $40,000. Under a standard $500,000 aggregate reserve ceiling, the solver approved the entire portfolio, since the leakage-free model's conservative PD estimates kept total portfolio Expected Loss (≈$119,324) well within budget — reinforcing the model's suitability as a cautious, at-origination screening tool. To verify the solver's constraint-handling mechanism under a materially tighter capital regime, a stress-tested $50,000 ceiling was also evaluated (Appendix C.4), under which the solver optimally rejected the 68 highest-risk applications while retaining 93.2% approval and 99.95% budget utilization[6]. This demonstrates the viability of linking LightGBM PD outputs directly to dynamic, constraint-driven capital reserving strategies, allowing an institution to programmatically balance risk appetite against revenue generation as the reserve ceiling tightens.

Scenario	Approved	Approval Rate	Total Expected Loss	Budget Utilization	Avg. PD (approved)
Pre-Optimized (approve all)	1,000 / 1,000	100.0%	$119,324	—	1.23%
Solver-Optimized ($500,000 cap)	1,000 / 1,000	100.0%	$119,324	23.9%	1.23%
Solver-Optimized ($50,000 cap, stress test)	932 / 1,000	93.2%	$49,975	99.9%	0.73%

Table 8: Solver-Optimized Expected Loss Portfolio Allocation

[6] The exact figures reported are reproducible given `numpy's default_rng(seed=42)`, but may differ marginally when run with different `scipy` or `imbalanced-learn` versions, consistent with the SMOTE version-sensitivity discussed in Section 6.4.

7.3.2. Cost-Sensitive Decision Optimization

While this study utilized a simulated threshold for demonstration purposes, future iterations must adopt a mathematically rigorous approach to threshold selection. By employing a cost-sensitive optimization function, lenders can dynamically adjust the decision boundary to integrate the bank's specific asymmetric cost matrix—weighing the missed lifetime value of a falsely rejected applicant against the realized financial loss of an unpredicted default. This ensures the model's operating point maximizes long-term portfolio profitability rather than purely statistical metrics.

7.3.3. Fairness and Demographic Validation

A primary limitation of the OpenIntro dataset is the absence of protected demographic attributes (e.g., race, gender, age), which restricted the depth of the Fair Lending audit to basic income proxies. Future implementations must utilize internal banking data containing these attributes to fully satisfy Equal Credit Opportunity Act (ECOA) requirements and conduct a definitive disparate impact audit.

7.3.4. Production Deployment and Continuous Monitoring

Transitioning this model from a static academic environment to a real-world banking production system introduces continuous model risk management challenges. Under SR 11-7 guidance, a deployed LightGBM model requires continuous monitoring for data drift (using Population Stability Index and Kolmogorov-Smirnov tests), particularly in volatile macroeconomic regimes. Furthermore, optimized inference pipelines must be engineered to ensure that computationally expensive SHAP adverse action explanations can be generated with millisecond latency.

7.3.5. Dataset Expansion, Temporal Validation, and Rigorous Evaluation

A central limitation of the present study is that the dataset of 10,000 observations is small relative to the high-dimensional complexity of the research question. Managing 55 raw variables that expand to 113 encoded features—combined with SMOTE resampling, multi-model ensemble training, SHAP decomposition, threshold optimization, and stress testing—means the sample size becomes constrained relative to the broad scope of inference, increasing the risk of overfitting and sample variance. Future research should evaluate the framework using substantially larger and temporally diverse credit portfolios containing a larger number of observed default events. External validation across different institutions would be particularly important for assessing generalizability and identifying distributional shifts that cannot be detected within the present dataset. With such dataset expansion, researchers can implement a more demanding evaluation pipeline—transitioning from a random train-test split to a strict Out-of-Time (OOT) temporal validation (i.e., training on historical loans to predict future loans). Because real-world

deployment requires predicting future borrower behavior under shifting macroeconomic conditions, temporal validation provides a far more realistic assessment of true generalization. Additionally, a fairer model-selection argument requires optimizing decision boundaries via stratified cross-validation within the training set across all models (Logistic Regression, Random Forest, XGBoost, and LightGBM), evaluating their true generalization using Precision-Recall AUC (PR-AUC) and the Matthews Correlation Coefficient (MCC) on the temporally segregated hold-out set. Finally, to account for the high variance and statistical uncertainty inherent in small minority-class sample sizes, future evaluations should incorporate bootstrap confidence intervals (e.g., 95% CIs) for all point estimates rather than relying solely on raw point values that can overstate precision.

7.4. Verdict on the 10,000-Record Challenge

In the Preface, this study posed a central question: could meaningful credit risk modelling and actionable predictions be extracted from a heavily constrained, immature 10,000-record snapshot?

The reported results illustrate different discrimination and operating-point trade-offs. Logistic regression identified 47.2% of positive cases at the standard threshold, but its precision was only 2.3%. LightGBM achieved a higher ROC AUC and was subsequently evaluated at an exploratory decision threshold. Furthermore, by deliberately optimizing the decision threshold to ~3.4%, the model achieved a precision lift that concentrated high-risk applicants nearly four times better than a random baseline. Most importantly, integrating SHAP proved that even under data scarcity, we could extract stable, localized risk drivers to satisfy adverse-action transparency.

However, the experiment also confirmed the professor's initial warnings regarding the ceiling of such a dataset. The immaturity of the outcome labels and the inherent probability distortion introduced by synthetic oversampling revealed that while a model can successfully rank risk under these conditions, its raw probability outputs cannot be safely calibrated for monetary Expected Loss calculations without a larger, fully matured historical cohort.

Ultimately, the 10,000-record challenge was a success not because it produced a flawless production system, but because it successfully illuminated the exact boundary between statistical classification and financial deployment. Having mapped these constraints, the theoretical and architectural foundations built in Part One now provide a mathematically honest, heavily tested blueprint ready to be scaled to a massive, out-of-time production environment in Part Two.

REFERENCES

Aboulmaouda, R., Slimani, K., & Chaoui, N. E. H. (2026). Credit risk scoring in the age of AI: A systematic comparison of traditional, ML, and DL models based on accuracy, stability, and interpretability. *Journal of Risk and Financial Management, 19*(7), 526. https://doi.org/10.3390/jrfm19070526

Alonso, A., & Carbó, J. M. (2020). *Machine learning in credit risk: Measuring the dilemma between prediction and supervisory cost* (Working Paper No. 2032). Banco de España. https://www.bde.es/f/webbde/SES/Secciones/Publicaciones/PublicacionesSeriadas/DocumentosTrabajo/20/Files/dt2032e.pdf

Altman, E. I. (1968). Financial ratios, discriminant analysis and the prediction of corporate bankruptcy. *The Journal of Finance, 23*(4), 589–609. https://doi.org/10.2307/2978933

Arain, S. U. R. (2026). *Machine Learning for Credit Risk Modeling: A Comparative Study of Traditional & Explainable AI Approaches* (Version 1.0) [Computer software]. https://github.com/sanaurrehmanarain/credit-risk-xai-framework

Basel Committee on Banking Supervision. (2010). *Basel III: A global regulatory framework for more resilient banks and banking systems*. Bank for International Settlements. https://www.bis.org/publ/bcbs189_dec2010.htm

Bazarbash, M. (2019). *FinTech in financial inclusion: Machine learning applications in assessing credit risk* (IMF Working Paper No. WP/19/109). International Monetary Fund. https://www.imf.org/-/media/files/publications/wp/2019/wpiea2019109.pdf

Bhuiyan, M., & Sweet, M. M. R. (2024). *Credit risk assessment using statistical and machine learning: Basic methodology and risk modeling applications*. SSRN. https://doi.org/10.2139/ssrn.5068722

Chawla, N. V., Bowyer, K. W., Hall, L. O., & Kegelmeyer, W. P. (2002). SMOTE: Synthetic minority over-sampling technique. *Journal of Artificial Intelligence Research, 16*, 321–357. https://doi.org/10.1613/jair.953

Chen, Z. (2024). *Machine learning in credit risk assessment: A comparative analysis of different models*. SSRN. https://doi.org/10.2139/ssrn.5006192

Dastile, X., Celik, T., & Potsane, M. (2020). Statistical and machine learning models in credit scoring: A systematic literature survey. *Applied Soft Computing, 91*, 106263. https://doi.org/10.1016/j.asoc.2020.106263

Federal Reserve. (2011). *Guidance on model risk management* (SR 11-7). Board of Governors of the Federal Reserve System. https://www.federalreserve.gov/boarddocs/srletters/2011/sr1107.pdf

Hadji Misheva, B., Osterrieder, J., Hirsa, A., Kulkarni, O., & Lin, S. F. (2021). *Explainable AI in credit risk management* [Preprint]. arXiv. https://arxiv.org/abs/2103.00949

Hardt, M., Price, E., & Srebro, N. (2016). Equality of opportunity in supervised learning. *Advances in Neural Information Processing Systems, 29*, 3315–3323. https://arxiv.org/abs/1610.02413

Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q., & Liu, T. (2017). LightGBM: A highly efficient gradient boosting decision tree. *Advances in Neural Information Processing Systems, 30*, 3146–3154. https://proceedings.neurips.cc/paper/2017/hash/6449f44a102fde848669bdd9eb6b76fa-Abstract.html

Khandani, A. E., Kim, A. J., & Lo, A. W. (2010). Consumer credit-risk models via machine-learning algorithms. *Journal of Banking & Finance, 34*(11), 2767–2787. https://doi.org/10.1016/j.jbankfin.2010.06.001

Lessmann, S., Baesens, B., Seow, H. V., & Thomas, L. C. (2015). Benchmarking state-of-the-art classification algorithms for credit scoring: An update of research. *European Journal of Operational Research, 247*(1), 124–136. https://doi.org/10.1016/j.ejor.2015.05.030

Lundberg, S. M., & Lee, S.-I. (2017). A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems, 30*. https://proceedings.neurips.cc/paper/2017/hash/8a20a8621978632d76c43dfd28b67767-Abstract.html

Montevechi, A. A., Miranda, R. de C., Medeiros, A. L., & Montevechi, J. A. B. (2024). Advancing credit risk modelling with machine learning: A comprehensive review of the state-of-the-art. *Engineering Applications of Artificial Intelligence, 137*, 109082. https://doi.org/10.1016/j.engappai.2024.109082

Noriega, J. P., Rivera, L. A., & Herrera, J. A. (2023). Machine learning for credit risk prediction: A systematic literature review. *Data, 8*(11), 169. https://doi.org/10.3390/data8110169

OpenIntro. (n.d.). *Loan data from Lending Club (loans_full_schema)* [Data set]. https://www.openintro.org/data/index.php?data=loans_full_schema

Pathi, S. P., & Pothineni, J. S. (2025). Interpretable AI in credit scoring: A comparative survey of SHAP, LIME, and hybrid approaches. *The American Journal of Engineering and Technology, 7*(11), 151–155. https://doi.org/10.37547/tajet/v7i11-304

SCALE D2C. (2026, April 13). *Explainable AI for financial decisions: SHAP for credit models*. https://scaled2c.com/blog/fintech-embedded-finance/explainable-ai-for-financial-decisions-shap-for-credit-models.html

Shi, S., Tse, R., Luo, W., D'Addona, S., & Pau, G. (2022). Machine learning-driven credit risk: A systemic review. *Neural Computing and Applications, 34*(17), 14327–14339. https://doi.org/10.1007/s00521-022-07472-2

Suhadolnik, N., Ueyama, J., & Da Silva, S. (2023). Machine learning for enhanced credit risk assessment: An empirical approach. *Journal of Risk and Financial Management, 16*(12), 496. https://doi.org/10.3390/jrfm16120496

APPENDICES:

Appendix A: Data Dictionary

Variable Name	Type	Description
`loan_amount`	Numerical	The listed amount of the loan applied for by the borrower.
`term`	Numerical	The number of monthly payments on the loan (36 or 60).
`interest_rate`	Numerical	Interest rate on the loan (%).
`installment`	Numerical	The monthly payment owed by the borrower.
`grade`	Categorical	Assigned loan grade (A = Safe to G = Risky).
`sub_grade`	Categorical	Finer-grained grade subdivision (dropped in feature engineering as redundant with `grade`).
`emp_length`	Numerical	Employment length in years (0–10, capped).
`homeownership`	Categorical	Home ownership status (RENT, OWN, MORTGAGE).
`annual_income`	Numerical	Self-reported annual income.

verified_income	Categorical	Indicates whether income was verified by the lender (Verified, Source Verified, Not Verified).
debt_to_income	Numerical	Ratio of total monthly debt payments to monthly income.
delinq_2y	Numerical	Number of 30+ day delinquencies in the borrower's credit file over the past 2 years.
total_credit_lines	Numerical	Total number of credit lines in the borrower's credit file.
months_since_last_delinq	Numerical	Months since the borrower's last delinquency (imputed to 999 where missing, indicating no delinquency history).
months_since_90d_late	Numerical	Months since a 90-day-late payment (imputed to 999 where missing).
state	Categorical	U.S. state of residence (50 levels, one-hot encoded).
loan_purpose	Categorical	Stated reason for the loan (e.g., debt consolidation, home improvement).
application_type	Categorical	Individual or joint application.
initial_listing_status	Categorical	Whether the loan was listed as whole or fractional.
disbursement_method	Categorical	Method of fund disbursement (Cash or Direct Pay).
loan_status_binary	Target	Binary target: 0 = Good (Current/Fully Paid), 1 = Risk (Late/In Grace Period/Charged Off).

Table A-1: Selected Variables from OpenIntro Loans Full Schema

Variable Name	Type	Reason for Exclusion
`loan_status`	Text (source)	Original loan status field (e.g., Current, Fully Paid, Charged Off); used to derive the binary target and dropped prior to modeling.
`balance`	Numerical	Post-origination repayment field — reflects current outstanding balance, observable only after disbursement.
`paid_total`	Numerical	Post-origination repayment field — total amount repaid to date.
`paid_principal`	Numerical	Post-origination repayment field — principal repaid to date; identified as the primary source of target leakage (Section 4.1.2).
`paid_interest`	Numerical	Post-origination repayment field — interest repaid to date.
`paid_late_fees`	Numerical	Post-origination repayment field — late fees accrued; mechanically correlated with default status.

Table A-2: Fields Excluded from the Predictor Set (Target Leakage)

Table A1 & Table A2: *Selected Variables from the OpenIntro Loans Full Schema Dataset, reflecting actual column names used in the processed dataframe (*`clean_data.csv`*). Post-origination fields are listed separately to make explicit which variables were excluded from the final model and why.*

Appendix B: Additional Figures

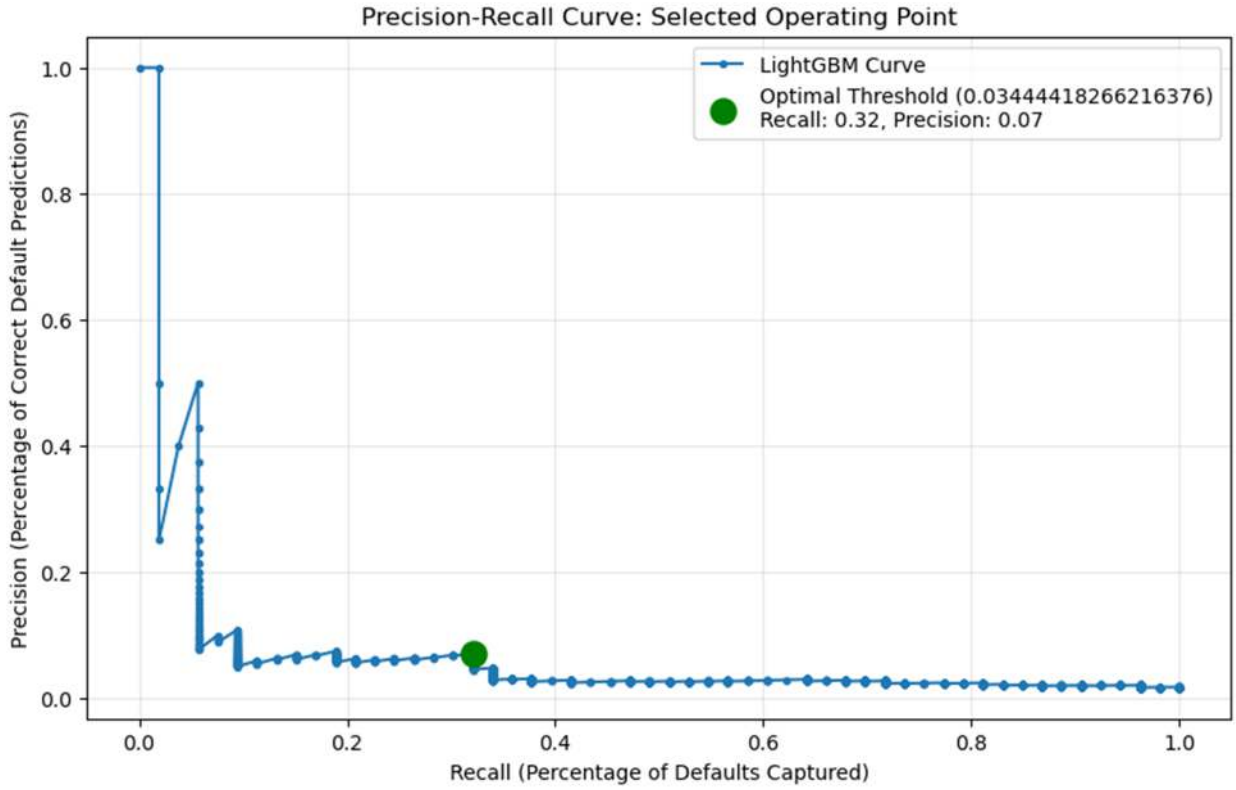


Figure B-1: Precision-Recall Curve and Threshold Optimization.

The curve illustrates the trade-off between Precision (vertical axis) and Recall (horizontal axis) for the LightGBM model. The green marker indicates the selected operating threshold of **~3.4%**. At this point, the model achieved a Precision of **7.1%** and a Recall of **32.1%**. This threshold represents the operating point used for the study, balancing Precision and Recall while providing a substantial improvement over the baseline default rate (Lift). It prioritizes reducing False Positives (thereby lowering unnecessary operational costs) while still identifying a meaningful proportion of the highest-risk borrowers. Although this threshold was selected on the hold-out test set for demonstration purposes, a production system should determine the operating threshold using a separate validation set or cross-validation to ensure an unbiased performance estimate.

Appendix C: Code Snippets

The code excerpts in Sections C.1–C.5[7] are drawn from Notebooks 03–08 and are presented in the same order in which they were executed. Each block depends on variables (`X`, `y`, `model`, etc.) established

[7] Appendix C Code Snippets are present in notebooks-&-codes/Appendix-C-Code-Snippets.ipynb.

in the preceding one, and the full sequence is intended to be run within a single kernel session, consistent with the accompanying notebook files.

C.1. SMOTE Implementation (Handling Imbalance)

Synthetic Minority Over-sampling Technique (SMOTE) was applied exclusively to the training partition, after the train/test split, to prevent synthetic samples from leaking into the hold-out evaluation set.

```
# From Notebook 03_feature_engineering.ipynb
import pandas as pd
from sklearn.model_selection import train_test_split
from imblearn.over_sampling import SMOTE
from collections import Counter

# 1. Load data and define Features (X) and Target (y)
# Post-origination leakage features ('paid_principal', 'paid_total', etc.)
# are dropped here, per the at-origination validation in Section 4.1.2.
df = pd.read_csv('../data/clean_data.csv')
df = df.drop(columns=['loan_status'])
df = df.drop(columns=[
    'sub_grade', 'issue_month', 'balance',
    'paid_total', 'paid_principal', 'paid_interest', 'paid_late_fees',
])

categorical_cols = [
    'state', 'homeownership', 'verified_income', 'verification_income_joint',
    'loan_purpose', 'application_type', 'grade',
    'initial_listing_status', 'disbursement_method',
]
df_encoded = pd.get_dummies(df, columns=categorical_cols, drop_first=True)

y = df_encoded['loan_status_binary']
X = df_encoded.drop(columns=['loan_status_binary'])
if X.isnull().sum().sum() > 0:
    X = X.fillna(X.median())

# 2. Stratified Train/Test Split (70/30)
# Stratification preserves the minority (default) class ratio in both
partitions.
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42, stratify=y
)

# 3. Apply Synthetic Minority Over-sampling Technique (SMOTE)
# Applied to the TRAINING partition only, to avoid leaking synthetic samples
# into the hold-out test set used for final evaluation.
print(f"Original Class Distribution: {Counter(y_train)}")
smote = SMOTE(random_state=42)
X_train_resampled, y_train_resampled = smote.fit_resample(X_train, y_train)
print(f"Resampled Class Distribution: {Counter(y_train_resampled)}")
# Result: Minority class (default) is balanced to match the majority class.

# 4. Persist processed data for downstream notebooks (04-08)
X_train_resampled.to_csv('../data/X_train_resampled.csv', index=False)
y_train_resampled.to_csv('../data/y_train_resampled.csv', index=False)
X_test.to_csv('../data/X_test.csv', index=False)
y_test.to_csv('../data/y_test.csv', index=False)
```

C.2. LightGBM Model Training

LightGBM was benchmarked against Logistic Regression, Random Forest, and XGBoost using identical SMOTE-resampled training data. LightGBM used library defaults except for the explicitly specified `random_state=42` and `verbose=-1` settings, per Table 2: LightGBM Configuration.

```
# From Notebook 04_model_training.ipynb
from lightgbm import LGBMClassifier

# Initialize Gradient Boosting Model with robust defaults
model = LGBMClassifier(
    random_state=42,
    verbose=-1
)

# Train on balanced data
model.fit(X_train_resampled, y_train_resampled)
```

C.3. SHAP Explainability

SHAP `TreeExplainer` was applied to the retrained LightGBM model to compute feature attributions in raw margin (log-odds) space, isolating Class 1 (default) contributions for both global and local interpretability analysis.

```
# From Notebook 05_explainability_analysis.ipynb
import numpy as np
import shap

# Load the SHAP JS visualization library into the notebook -- required once
# per kernel session, otherwise force_plot() below silently renders a blank
# "Visualization omitted" box even though it executes without error.
shap.initjs()

# Initialize Explainer
explainer = shap.TreeExplainer(model)
shap_values = explainer.shap_values(X_test)

# Newer SHAP versions return a single array for binary classifiers instead of
# a [class0, class1] list -- this guard keeps the snippet version-safe.
if isinstance(shap_values, list):
    shap_values = shap_values[1]

base_value = explainer.expected_value
if isinstance(base_value, (list, np.ndarray)):
    base_value = base_value[1]

# Generate Force Plot for Local Interpretation (Borrower #1)
shap.force_plot(
    base_value,
    shap_values[0, :],
    X_test.iloc[0, :]
)
```

C.4. Expected Loss Optimization Solver

The following script formulates and solves the binary linear program used to generate Table 8 in Chapter 7, using `scipy.optimize.milp`. Decision variables $x_i \in \{0, 1\}$ indicate loan approval; the objective maximizes $\sum x_i$ subject to $\sum \text{EL}_i \, x_i \leq \text{budget}$, where $\text{EL}_i = \text{PD}_i \times \text{LGD} \times \text{EAD}_i$. PD values are drawn from the trained LightGBM model's real test-set predictions (reproducing notebooks 03–04), rather than simulated, so results reflect the model's actual risk profile rather than an idealized one.

```
import numpy as np
import pandas as pd
from scipy.optimize import Bounds, LinearConstraint, milp

RANDOM_STATE = 42

# ---------------------------------------------------------------
# 1. Real PD estimates from the model trained in C.2, on the same
#    hold-out test set constructed in C.1
# ---------------------------------------------------------------
pd_test = model.predict_proba(X_test)[:, 1]

# ---------------------------------------------------------------
# 2. Build the simulated 1,000-loan portfolio
# ---------------------------------------------------------------
rng = np.random.default_rng(RANDOM_STATE)
N_LOANS = 1000
LGD = 0.45

sample_idx = rng.choice(len(pd_test), size=N_LOANS, replace=False)
PD = pd_test[sample_idx]
EAD = rng.uniform(1_000.0, 40_000.0, size=N_LOANS)
EL = PD * LGD * EAD

portfolio = pd.DataFrame({
    "Loan_ID": np.arange(1, N_LOANS + 1),
    "PD": PD, "LGD": LGD, "EAD": EAD, "Expected_Loss": EL,
})
```

```
# ------------------------------------------------------------------
# 3. Solve: maximize sum(x_i) s.t. sum(EL_i * x_i) <= budget, x_i in {0,1}
# ------------------------------------------------------------------
def solve_portfolio(expected_loss: np.ndarray, budget: float):
    n = len(expected_loss)
    objective = -np.ones(n)  # milp minimizes -> negate to maximize approvals
    constraint = LinearConstraint(expected_loss.reshape(1, -1), lb=-np.inf,
ub=budget)
    bounds = Bounds(lb=0, ub=1)
    result = milp(c=objective, constraints=constraint, bounds=bounds,
                  integrality=np.ones(n), options={"time_limit": 300})
    if not result.success:
        raise RuntimeError(f"Optimization failed: {result.message}")
    return np.rint(result.x).astype(int)


def summarize(approved: np.ndarray, budget: float, label: str) -> dict:
    mask = approved == 1
    total_el = EL[mask].sum()
    return {
        "Scenario": label,
        "Budget": budget,
        "Approved": int(mask.sum()),
        "Approval Rate": mask.sum() / N_LOANS,
        "Total Expected Loss": total_el,
        "Budget Utilization": total_el / budget if budget else np.nan,
        "Avg PD (approved)": PD[mask].mean(),
        "Avg EAD (approved)": EAD[mask].mean(),
    }


# Baseline: naive full approval, no risk-based screening
baseline_approved = np.ones(N_LOANS, dtype=int)

# Chapter 7 headline scenario: the specified $500,000 budget
approved_500k = solve_portfolio(EL, 500_000)

# Appendix stress test: a materially tighter capital ceiling
```

```
approved_50k = solve_portfolio(EL, 50_000)

results = pd.DataFrame([
    summarize(baseline_approved, np.inf, "Pre-Optimized (approve all)"),
    summarize(approved_500k, 500_000, "Solver-Optimized ($500,000 cap)"),
    summarize(approved_50k, 50_000, "Solver-Optimized ($50,000 cap, stress
test)"),
])

pd.set_option("display.float_format", lambda v: f"{v:,.4f}")
print(results.to_string(index=False))

# ------------------------------------------------------------------
# 4. Appendix sensitivity sweep across multiple capital ceilings
# ------------------------------------------------------------------
sweep_budgets = [500_000, 100_000, 50_000, 30_000, 15_000, 5_000]
sweep_rows = []
for b in sweep_budgets:
    appr = solve_portfolio(EL, b)
    sweep_rows.append(summarize(appr, b, f"${b:,.0f} cap"))
sweep_df = pd.DataFrame(sweep_rows)
print("\nSensitivity sweep:")
print(sweep_df.to_string(index=False))
```

Capital Ceiling	Approved	Approval Rate	Total Expected Loss	Budget Utilization
$500,000	1,000 / 1,000	100.0%	$119,324	23.9%
$100,000	996 / 1,000	99.6%	$98,565	98.6%
$50,000	932 / 1,000	93.2%	$49,975	99.9%
$30,000	847 / 1,000	84.7%	$29,846	99.5%
$15,000	703 / 1,000	70.3%	$14,936	99.6%
$5,000	462 / 1,000	46.2%	$4,984	99.7%

Table C-1: Extended sensitivity table. (Shows the mechanism binding across multiple reserve ceilings).

C.5. Adversarial Stress Test

To evaluate model risk under extreme, out-of-distribution conditions per SR 11-7 guidance, 50 adversarial borrower profiles were generated by applying compounding stress to a real borrower sample, then scored with the trained LightGBM model to produce Table 4 in Section 6.3.

```
# From Notebook 08_Threshold_Optimization.ipynb (extension)
# 0. Baseline in-distribution PD range (for comparison against adversarial
results)
baseline_pd = model.predict_proba(X_test)[:, 1]
print(f"In-distribution PD range: [{baseline_pd.min():.4f},
{baseline_pd.max():.4f}]")

# Adversarial Stress Test: Generation and Scoring
# Reuses the LightGBM model trained in C.2 and the categorical_cols /
# cols_to_drop encoding scheme established in C.1.
import numpy as np
import pandas as pd

# 1. Load 50 real borrower rows as a realistic baseline for the ~40 raw
#    attributes not deliberately stressed (credit history, account counts,
#    etc.), preserving realistic feature correlations that a fully synthetic
#    row generator could not guarantee.
raw = pd.read_csv('../data/clean_data.csv')
base_rows = raw.drop(columns=['loan_status']).sample(n=50,
random_state=42).reset_index(drop=True)

# 2. Apply compounding macroeconomic stress to the shock dimensions only:
#    interest rate and debt-to-income pushed beyond anything seen in
#    training, combined with reduced income and increased exposure.
rng = np.random.default_rng(42)
adv = base_rows.copy()
adv['interest_rate']   = rng.uniform(25.01, 40.0, size=50)    # spike above 25%
adv['debt_to_income'] = rng.uniform(100.01, 180.0, size=50) # above 100
adv['annual_income']   = rng.uniform(15000, 30000, size=50)   # compounding:
depressed income
adv['loan_amount']     = rng.uniform(30000, 40000, size=50)   # compounding:
high exposure
adv['term']            = rng.choice([36, 60], size=50)
```

```
# 3. Recompute 'installment' using the standard fixed-rate amortization
#    formula (NOT a rough heuristic) so the stressed installment values
#    are internally consistent with the shocked rate/amount/term:
#      installment = P * (r/12) / (1 - (1 + r/12)^-n)
monthly_rate = (adv['interest_rate'] / 100) / 12
adv['installment'] = adv['loan_amount'] * monthly_rate / (1 - (1 +
monthly_rate) ** (-adv['term']))

# 4. Reproduce the exact leakage-free encoding pipeline (C.1) so
#    the adversarial rows are scoreable by the model trained in C.2.
cols_to_drop = ['sub_grade', 'issue_month', 'balance', 'paid_total',
                'paid_principal', 'paid_interest', 'paid_late_fees']
# (categorical_cols already defined in C.1 -- reused here unchanged)

adv_clean = adv.drop(columns=[c for c in cols_to_drop if c in adv.columns])
adv_encoded = pd.get_dummies(adv_clean, columns=categorical_cols,
drop_first=True)
adv_encoded = adv_encoded.reindex(columns=model.feature_name_, fill_value=0)

# 5. Score the adversarial portfolio with the trained LightGBM model
adv_pd = model.predict_proba(adv_encoded)[:, 1]
adv['predicted_PD'] = adv_pd

print(f"Adversarial PD range: [{adv_pd.min():.4f}, {adv_pd.max():.4f}]")
print(f"Any PD >= 0.99 (erratic output)?  {(adv_pd >= 0.99).any()}")

# 6. Extract the 5 most extreme cases for Table 4
extreme_5 = adv.sort_values('predicted_PD', ascending=False).head(5)
cols = ['loan_amount', 'term', 'interest_rate', 'installment',
        'annual_income', 'debt_to_income', 'predicted_PD']
print(extreme_5[cols].round(3).to_string(index=False))
```

MACHINE LEARNING FOR CREDIT RISK MODELING

A comparative study of traditional and explainable AI approaches

THESIS AT A GLANCE

QUESTION

Can explainable ML improve minority-class default detection without sacrificing transparency?

DATA

OpenIntro Loans Full Schema: 10,000 loans, 55 variables, 1.7% default rate.

DESIGN

70/30 hold-out split; random seed 42; SMOTE applied only to training data.

METHODS PIPELINE

1 Select pre-origination variables → 2 Split train / test → 3 SMOTE training only → 4 Train 4 models → 5 Tune threshold → 6 Explain + audit

EXPERIMENTS AND RESULTS

MODEL COMPARISON

Four candidates: Logistic Regression, Random Forest, XGBoost, and LightGBM. Random Forest had the highest global ROC AUC, but it collapsed to the majority class at threshold 0.50. LightGBM retained useful minority-class sensitivity.

SELECTED OPERATING POINT

LightGBM was selected for practical default detection. At the exploratory threshold ~3.4%, Precision = 7.1% and Recall = 32.1%. These figures are optimistic because the hold-out test set was used for threshold calibration.

0.670	0.659	7.1%	32.1%
best ROC AUC	selected ROC AUC	precision @ ~3.4%	recall @ ~3.4%

EXPLAINABILITY

SHAP identified the main risk drivers as interest rate, debt-to-income ratio, and loan grade. This connects model predictions to borrower-level and global explanations.

FAIRNESS CHECK

Using income as a proxy and False Positive Rate parity: low-income FPR = 0.07%; high-income FPR = 0.13%. No evidence of disparate impact was found under the evaluated metric.

CONCLUSIONS AND LIMITATIONS

TAKEAWAY

A constrained LightGBM framework can provide useful minority-class discrimination while remaining interpretable through SHAP. Deployment requires leakage controls, validation-based threshold selection, monitoring, and governance.

LIMITATIONS

Single dataset; severe imbalance; income used only as a proxy; test-set threshold calibration introduces optimistic bias; results are not a definitive deployment estimate.

Reproducibility: Python 3.13.5 | scikit-learn 1.8 | LightGBM 4.7 | imbalanced-learn 0.11 | seed = 42

Sana Ur Rehman Arain | 2026